

Counterfactual modeling using deep learning reveals spatially-explicit impact of protected areas in the Amazon forests

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Abstract

Tropical forests are under threat, with dire consequences for biodiversity and climate change. Protected areas are the most common approach to forest conservation, but their successful expansion depends on identifying which protected areas have reduced deforestation and where. Since the factors that drive deforestation are highly non-linear and location-specific, to measure the effectiveness of protection, we need an approach to evaluation that can capture heterogeneity across space and time. We introduce a physics-constrained, multimodal deep learning (DL) model that predicts deforestation that would have occurred in the absence of protection, learning from deforestation drivers and observed deforestation in unprotected forest, including satellite image embeddings from a vision foundation model. Applied across the Amazon basin from 1996 to 2020, the DL model generates spatially explicit counterfactuals at 1 km resolution, providing the first gridcell-level estimates of individual protected area effectiveness. Comparing these counterfactuals to observed deforestation, we resolve not only which protected areas worked on average, but we also disentangle effectiveness within each protected area, heterogeneity that aggregate methods do not capture. We find that protection was strongest in frontier regions under the highest agricultural pressure. We further find our estimates of average effectiveness are roughly half the size of standard econometric difference-in-differences model estimates. Because the DL model learns deforestation dynamics rather than rules specific to the Amazon, the same approach can map conservation impact in other regions, supporting spatially targeted action under global targets such as 30×30.

1 Introduction

The Amazon is home to the world’s largest tropical forest, storing immense amounts of carbon and biodiversity, and regulating the global climate system. Aboveground vegetation biomass in the Amazon basin is estimated to hold 93–120 PgC (Baccini et al., 2012; Malhi et al., 2008), while soils and below-ground biomass add substantially to this stock (Saatchi et al., 2011; Walker et al., 2020). The basin also harbors unparalleled biodiversity, supporting more than 10% of the world’s known species (Dirzo & Raven, 2003; Malhi et al., 2008) and plays a critical role in sustaining rainfall through land–atmosphere feedback, (Nobre et al., 2016; Spracklen & Garcia-Carreras, 2015)

Yet this global asset is under threat. The Amazon faces mounting pressures from agriculture, forestry, fires, urbanization, and mining (Curtis et al., 2018). Already, one fifth of the Amazon biome has been deforested, and if current rates continue, 27% could be lost by 2030. Deforestation and forest degradation have resulted in gross emissions in the order of 0.5 PgC annually in recent decades, undermining the Amazon’s role as a carbon sink (Baccini et al., 2017; Harris et al., 2021). Climate–land use interactions further amplify risks, with projections suggesting that continued deforestation could push the system toward a tipping point of large-scale forest dieback (Nobre et al., 2016; Walker et al., 2020).

In response, governments have protected large portions of the Amazon, with formal protected areas now covering roughly a quarter of the basin (Amazonian Network of Georeferenced Socio-Environmental Information (RAISG), 2023).¹ Globally, protected areas are the most widely used form of conservation

¹Indigenous lands cover a further 27% of the basin (Amazonian Network of Georeferenced Socio-Environmental Information (RAISG), 2023). This paper focuses exclusively on protected areas, and cells with other conservation designations are excluded from the control pool (see Methods).

intervention. These interventions have been found to slow deforestation in some regions (Andam et al., 2008; Nolte et al., 2013), but their effectiveness varies with governance regime and local context (Abman, 2018; Shah et al., 2021). For example, strictly protected areas have on average performed better at avoiding deforestation than sustainable-use reserves, while Indigenous lands have shown comparable or stronger performance in certain frontier regions (Reynaert et al., 2024).

The effectiveness of conservation policies has been evaluated at national or subnational scales, but consistent Amazon basin-wide assessments remain scarce, limiting our ability to target future conservation efforts. (Andam et al., 2008; Nolte et al., 2013; Pfaff et al., 2015). Evaluating PA effectiveness is inherently challenging because deforestation dynamics are highly non-linear, shaped by heterogeneous socio-environmental drivers, and are subject to strong spatial and temporal dependencies. Traditional econometric models often struggle to capture these complexities, relying on rigid functional forms and may not capture the underlying heterogeneity (Shah et al., 2021). In this paper, we develop a physics-constrained, multimodal deep learning (DL) model to estimate the effect of protected areas across the Amazon from 1996–2020 for each protected 1 km².

Deep learning models flexibly represent non-linear processes and spatial heterogeneity without pre-specified forms, making them well suited to the scale and complexity of Amazonian deforestation (Mullainathan & Spiess, 2017). Recent advances in satellite monitoring, open-access land-cover data, and artificial intelligence make it possible to train models that learn deforestation dynamics at great spatial detail. Here we train a DL model on unprotected forest observations and generate counterfactual deforestation predictions representing what would have occurred in the absence of protection. The DL model integrates temporal deforestation trends, geographic covariates, spatial context, and fine-scale landscape information derived from satellite imagery. The model both predicts whether and when a gridcell will be deforested, and by what amount, allowing us to estimate the total quantity of deforestation avoided by protection. Constraints built into the DL model ensure that model predictions respect basic ecological limits at every step, yielding physically plausible counterfactual trajectories across the full prediction horizon. To validate the causal interpretation, we benchmark against staggered adoption difference-in-differences models, enabling direct comparison between the two approaches.

Prior evaluations of protected areas have largely used matching, often combined with differences in differences, to address potential selection bias driven by the placement of protected areas (Abman, 2018; Andam et al., 2008; Ferraro et al., 2013; Geldmann et al., 2019; Joppa & Pfaff, 2009, 2010, 2011; Keles et al., 2023; Pfaff et al., 2015; Reynaert et al., 2024; Shah et al., 2021; Sims & Alix-Garcia, 2017), including the use of synthetic controls (Sills et al., 2015). Addressing selection is critical for estimating the effect of protection: early work on the Amazon demonstrates that taking selection into account reduces the estimates of avoided deforestation by 4% to 2% (Pfaff et al., 2015). Their work along with other work evaluating protected areas in Brazil demonstrates that the effectiveness of PAs varies by governance and by threat (Nolte et al., 2013). A couple recent papers take the staggered introduction of PAs into account, allowing for treatment heterogeneity over time (Grupp et al., 2026; Rico-Straffon et al., 2023). Two recent papers apply more flexible approaches to estimate spatial heterogeneous effects of protection (Grupp et al., 2026; Nakamura et al., 2026), Double ML (Sanford, 2025) but these papers do not address the temporal dynamics that might be needed to evaluate a large number of PAs implemented over many years.

In this paper, we explicitly develop a model to predict what would have happened in absence of protection. Predicting counterfactual deforestation faces challenging decisions regarding what outcome to predict, at what spatial scale, and how far in the future. We build on a literature that predicts deforestation. Most of these papers use machine learning (ML) to predict the probability of a pixel being deforested in the following year, (Ball et al., 2022; Engelmann & Toetzke, 2023; Overlan et al., 2025).² While the binary construction comes from the deforestation data itself, the use of a binary classification has two implications. First, it complicates estimating how much forest would be cut (or carbon would be released) since the discrete measure does not easily translate to the volume of trees (or carbon) removed (Mayfield et al., 2017). Second, the models treat the change from forest to non-forest as permanent, so a pixel enters the sample once and leaves it as soon as it is deforested (Ball et al., 2022; Engelmann & Toetzke, 2023; Overlan et al., 2025). This data structure means that errors that result in a ‘false positive’, incorrectly suggesting a pixel will be deforested, will mechanically perpetuate over time, since that observation will not appear in future forecasts of potential deforestation. Some papers model a continuous measure of deforestation by averaging the predicted probability of deforestation to a larger

²A third literature detects deforestation from satellite imagery after it has occurred, either mapping the loss of forest directly (Torres et al., 2021) or identifying the land use that replaced the forest (Sbouei et al., 2023). Since we are specifically interested in models that use past data to forecast deforestation, we do not discuss this literature in detail.

gridcell. For example, Overlan et al. (2025) obtain a continuous measure of deforestation by averaging the predicted probabilities of deforestation of all pixels within tiles of 7.7 km. A tile of this size covers close to 60 km², which is larger than more than 200 of the protected areas in our sample.

Given that deforestation in a location is highly correlated over time and that protection may take several years to have its full effect, ideally a counterfactual would include not just a single year, but a predicted trajectory of deforestation in the absence of protection. How far ahead a forecast can predict is the third constraint, and no study in this literature tests more than a single year. A model trained to predict one year ahead can in principle be run repeatedly to reach later years, but how reliable those multi-year predictions are has not been tested (Overlan et al., 2025), and Ball et al. (2022) suggest that predicting further ahead would require changes to the model architecture, such as adding transformer components. Closest to what impact evaluation needs are Sales et al. (2017), who model the proportion of a 1 km grid cell deforested each year through a two-stage hurdle specification, the first stage predicting whether any deforestation occurs and the second predicting how much, with remaining forest as the strongest predictor of that amount. Their outcome is therefore continuous, and it can be reduced year after year, at the same spatial scale at which protection is evaluated. That model is nonetheless fit one year ahead and described by its authors as suited to two or three, so it gives no year-by-year path of deforestation across the period over which a protected area would need to be assessed. Ye et al. (2019) reach four years with a recurrent network, but supply it with observed input data rather than its own predictions throughout. Further, note that the predicted counterfactual trajectory needed for impact evaluation should not be trained on deforestation outcomes in locations after they are protected. If protection is effective, it will decrease deforestation, and using this post-treatment information would likely depress the predicted counterfactual deforestation in future years. Thus, a model must instead take its own prediction for one year as the input for the next, and continue in this way from the year protection began until the end of the evaluation period. Predicting in this manner introduces a difficulty that a single-year forecast never meets, because a small error in each year is carried into the following year’s inputs and accumulates, so that the predicted path of deforestation drifts steadily away from what would have occurred. None of the studies above predicts in this way, so none measures how far such a prediction drifts, none corrects for it, and none demonstrates that a multi-year counterfactual still matches deforestation in places where the true outcome is known. Where these studies say anything about protection, they do so through covariate coefficients and distance relationships rather than through causal identification (Larrea-Gallegos & Vázquez-Rowe, 2022; Sales et al., 2017), and none is trained to predict the deforestation that would have occurred in an area had it not been protected.

Closest in spirit to our model is a recent paper that uses random forest survival to predict deforestation pressure across the Amazon to inform placement of future PAs, instead of our approach where we use the predictions to estimate the effect of PAs ex-post (Inkinen et al., 2026). In our approach, we specifically train a model to forecast the amount of deforestation, in locations that are similar to protected areas, allowing us to use these predictions as an estimate of deforestation in absence of protection.

Beyond producing a basin-wide average, our framework generates a spatially explicit counterfactual map at 1 km resolution that can be interrogated at the level of individual protected areas. This capacity distinguishes our approach from traditional econometric methods, which estimate average treatment effects across matched units but cannot produce a mappable counterfactual. A conservation planner can compare observed deforestation in any specific PA against its predicted counterfactual and arrive at a credible gridcell-level estimate of how much forest that intervention saved. This resolution of spatial heterogeneity is the central contribution of the paper, and one that opens a new frontier in how conservation effectiveness is evaluated and communicated to policy makers.

2 Results

2.1 Deep learning model performance

We train a two-step DL model on unprotected forest gridcells and pre-protection years, and use it to generate counterfactual deforestation predictions for PA gridcell across a ± 10 -year window centered on the year of PA implementation. From $T = 0$ onward, predictions are generated autoregressively, as described in Methods, to avoid contamination from post-treatment observations where deforestation is already suppressed by protection.

We first assess model performance by comparing predictions with observed deforestation. Predicting deforestation is challenging because forest loss remains rare in relation to total forest cover, is spatially clustered, and driven by processes that vary sharply across contexts. To address the fact that deforestation

tion at a small spatial scale has a mass point at zero, our model follows a two-step procedure: initially the classification stage of the DL model predicts the probability of deforestation in the 1km gridcell, and then, if the predicted probability is greater than zero, the regression stage predicts the amount of deforestation in that same gridcell, otherwise the prediction remains zero. We evaluate the performance of both steps.

We assess our model against a held-out test set of over 460,000 observations (Table 1). The classification stage achieves an F1 score of 0.897 and a ROC AUC of 0.869, reflecting strong balance between precision and recall. The model is designed to prioritize recall over precision, given that false positives are corrected downstream by the regression stage. To contextualize performance of the DL model, we compute these metrics for the econometric model as well (see Appendix L).

The regression stage achieves an RMSE of 4.74% and MAE of 1.48% at 1 km annual resolution. The R^2 of 0.18 at this resolution is expected, because gridcell-level deforestation is inherently noisy and individual predictions carry high variance. At 10 km resolution, at which spatially independent noise averages out across gridcells, Pearson’s correlation rises from 0.44 to 0.74 and R^2 increases from 0.18 to 0.55, confirming that the model captures genuine spatial and temporal structure in forest loss rather than fitting to noise.

Table 1: Deep learning model performance on the held-out test set. The test set contains 462,216 cell-years at 1 km, aggregated to 75,643 blocks at 10 km.

Confusion matrix (%)			Classification metrics				
Actual	Predicted		Precision	Recall	F1 score	Accuracy	ROC AUC
	0	1	0.845	0.956	0.897	0.878	0.963
0	34.74	9.73	Regression metrics				
1	2.42	53.10					
Resolution	RMSE (%)		MAE (%)	R^2	Correlation		
1 km	4.74%		1.48%	0.18	0.44		
10 km	0.27%		0.06%	0.55	0.74		

Note: RMSE and MAE are percentages of cell area and so are not comparable across resolutions, unlike R^2 and correlation. Classification metrics use a 0.5 threshold; zero-outcome cells were downsampled to a 1:1 ratio before the train/test split.

Visually, DL predicted deforestation captures the spatial patterns of observed deforestation well for non-protected gridcells for the time period between 1996 and 2020 (see Appendix O). Aggregate deforestation and spatial patterns align nicely, despite the fact that the DL model is trained on selected gridcells that become protected later on and their matched counterparts – rather than the full set of forested gridcells in the Amazon biome.

Overall, the model predicts held-out deforestation accurately, providing a reliable basis for counterfactual estimation (further details on model training and performance are in Appendix K).

2.2 Counterfactual deforestation prediction

We generate counterfactual deforestation predictions for PA gridcells before and after their respective year of PA implementation between 1996-2020 and compare the prediction to observed deforestation. The difference between observed and predicted deforestation after PA implementation captures the avoided deforestation due to protection.

To confirm that divergence between predicted and observed deforestation reflects the genuine conservation impact rather than model overprediction patterns, we apply the same procedure to matched pseudo-PA gridcells, which are unprotected locations with similar pre-treatment characteristics to PA gridcells. Predicted and observed deforestation remain tightly aligned across the full ± 10 -year window for these control gridcells (Figure 1, left), confirming that the model is unbiased and that any spurious drift due to autoregressive prediction is taken care of by the bias correction step.

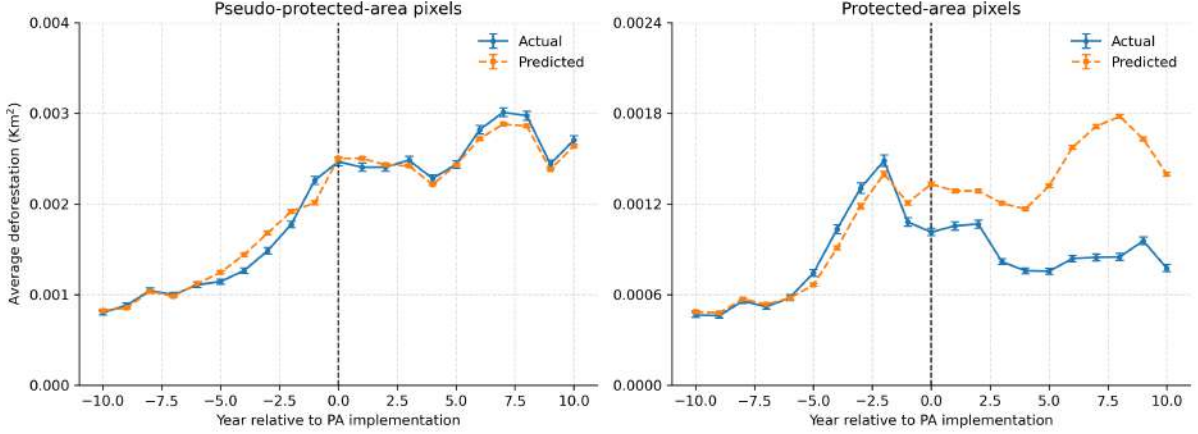


Figure 1: Counterfactual deforestation predictions for pseudo-protected-area gridcells (left) and protected-area gridcells (right) across a ± 10 -year window centered on PA implementation ($T = 0$, dashed line). Lines show observed deforestation (Actual) and the model’s counterfactual prediction (Predicted); error bars are 95% confidence intervals. For pseudo-PA gridcells the two lines overlap throughout the window. For PA gridcells they track closely before implementation and diverge from $T = 0$ onward, with observed deforestation falling below the counterfactual; this gap is the avoided deforestation attributable to protection.

Applied to actual PA gridcells, predicted and observed deforestation track closely through the pre-treatment period and diverge sharply from $T = 0$ onward, when observed deforestation falls well below the counterfactual (Figure 1, right). This gap represents avoided deforestation attributable to protection, which we quantify formally in the following sections.

2.3 Average impacts of protected areas on deforestation

We use the predicted counterfactuals in an econometric approach to evaluate the impact of PAs as averaged treatment effect on the treated (ATT). Year-by-country fixed effects are used on all regressions to account for institutional and political changes over time and space. For gridcells that become PAs eventually, we find that deforestation is reduced by 0.05% of the cell area, which equals 0.05 hectare of land, after their protection status is assigned. This finding is robust across various specifications as shown in Figure 2. The estimated impact is found when comparing the difference of predicted and observed deforestation (residuals) from the DL model before and after protection (PA DL residuals). It is consistent, when extending the econometric approach to a difference-in-differences (DiD) design by adding information of the pseudo-PAs (DID DL residuals) and to further including gridcell fixed effects (DID FE DL residuals). Since PAs are not implemented all in the same year, but over time, we further use an estimator accounting for the staggered nature of the treatment (Callaway & Sant’Anna, 2021) and show that the aggregate effect remains robust when averaging over the different treatment cohorts (DID staggered DL residuals). In contrast, running the simple before and after treatment model on our control set of pseudo-PAs (PPA DL residuals), does not suggest any significant change in deforestation. Furthermore, we compare the DL-based residual approach to a standard econometric approach directly applied to the matched data. Potential agricultural revenue is added as additional time-varying control variable. The simple DID approach (DID FE econometric defor.) as well as accounting for staggered treatment (DID staggered econometric defor.) result in significant, but twice as large treatment effects compared to our main results. According to these models, establishing protected areas has reduced deforestation on average by 0.1% of the cell area, which equals 0.1 hectare of land. This suggests that the standard econometric approach might overestimate the conservation impact of PAs, as it is less capable of accounting for spatial heterogeneity in counterfactual deforestation.

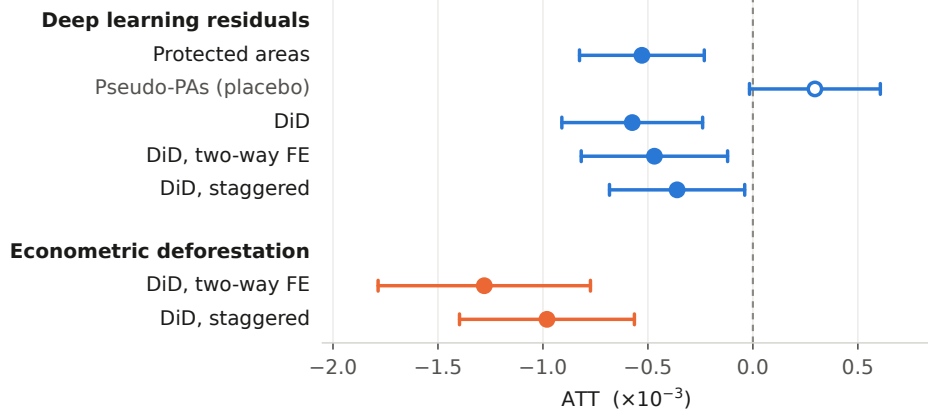


Figure 2: Average treatment effects on the treated (ATT). Estimation results also shown in Table Q1.

Figure 3 shows an event study plot based on the two DID models accounting for staggered treatment adoption. Both models support the assumption of parallel trends prior to treatment. Also, both figures suggest that avoided deforestation increased with longer protection status. Consistent with the ATT, the residual-based approach suggests a constantly smaller treatment effect over time than the econometric estimation directly on the matched data.

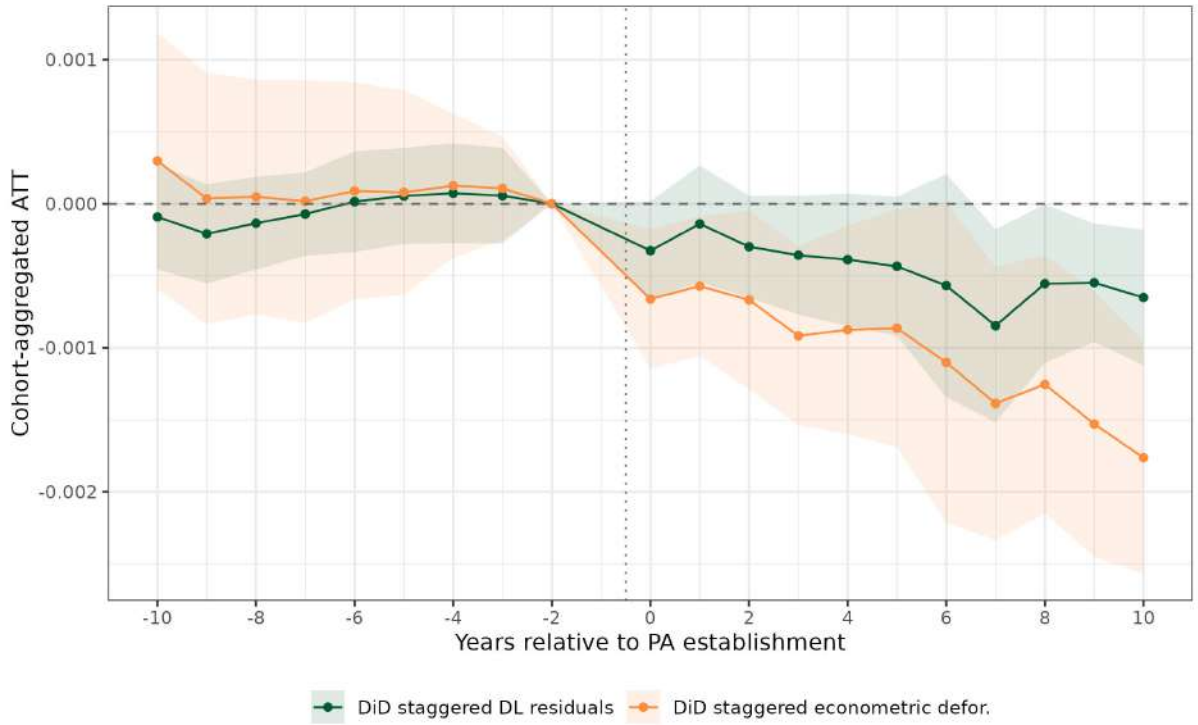


Figure 3: Event-study plots for staggered DiD models. Event-study plots for additional models are shown in Figure Q2.

2.4 Spatial heterogeneity in protected area impacts on deforestation

The spatially explicit counterfactual predictions of the DL model reveal where protection made the greatest difference in deforestation. Figure 4 maps the difference between predicted and observed deforestation across PA and pseudo-PA gridcells in the five years following implementation at 1 km resolution. Across the Amazon basin, conservation gains are spatially concentrated, with the largest effects in frontier regions, particularly along the southern and eastern arc of deforestation where agricultural and extractive

220 pressures are highest. In these areas, the model predicts substantial forest loss in the absence of protec-
221 tion, and observed deforestation falls well below those predictions, pointing to a strong protective effect
222 precisely where it is most needed. A basin-wide estimate of total avoided deforestation over the five-year
223 window following PA implementation is reported as 1,805 km². In some areas, however, observed defor-
224 estation exceeded model predictions, suggesting local governance failures, displacement of deforestation
225 pressure to adjacent unprotected land (leakage), or contexts where local drivers are not fully captured
226 by the model.

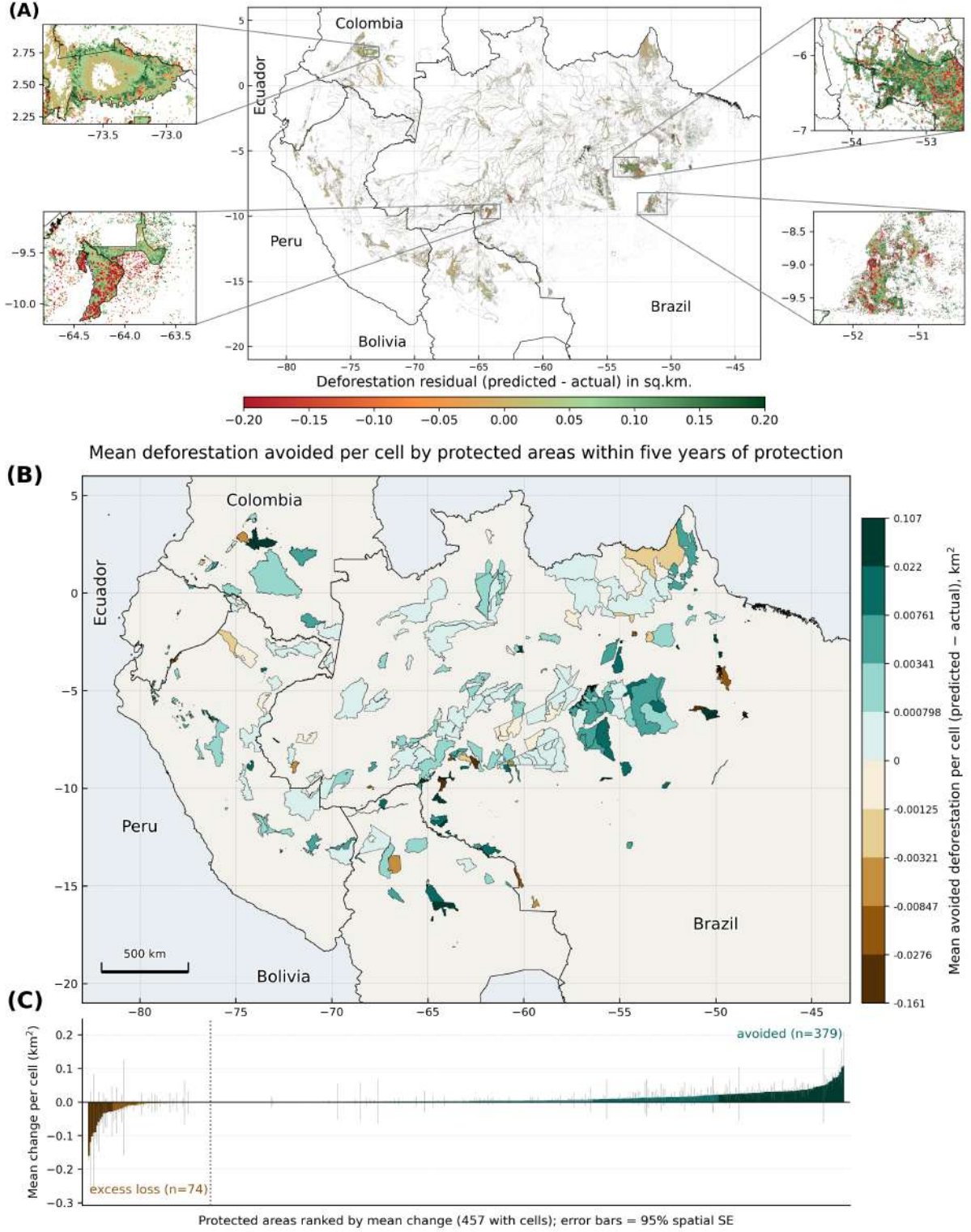


Figure 4: Spatially explicit avoided deforestation from protected areas over the five years following implementation ($T = 0$ to $T = 5$). **(A)** Gridcell-level difference between predicted counterfactual and observed deforestation (Predicted – Observed) at 1 km resolution. Gridcells where both predicted and observed deforestation were zero, and areas outside the matched sample, are masked in white. **(B)** The same residuals aggregated to individual protected-area polygons (teal = avoided deforestation, brown = excess forest loss). **(C)** Protected areas ranked by total effect ($|\text{change}| \geq 1 \text{ km}^2$).

3 Discussion and conclusions

Both the DiD model based on DL-generated counterfactuals and the standard econometric DiD model showed effective reduction of deforestation after implementation of protected areas in the Amazon. Only the staggered DiD on the matched data does not conclude with a significant change. Due to computational limitations, this estimation was run on small random subsample. Future efforts computing a larger sample that captures more variation could assess in how far this result is driven by the sampling.

The DL approach captured the fine-scale spatial variation in protection effectiveness. Despite that our DL model was trained specifically to predict deforestation in protected areas, its overall prediction performance for the Amazon basin captures the spatial deforestation of deforestation well. These comparisons show how the DL model can capture nonlinear and heterogeneous drivers of land-use change. Nonetheless, for answering a different research question, an adjusted, context-specific design of the DL model might be more suitable.

Our spatially explicit approach shows that in some areas, the establishment of protected areas has prevented the loss of completely forested cells. This is especially the case in frontier regions facing high deforestation pressure. At the same time, we find that deforestation pressure is low in most of the protected area cells and only small shares of deforestation are prevented in most of the protected cells. While this supports the often discussed concern that protected areas are not necessarily implemented in areas with the highest deforestation pressures, also the smaller contributions of avoided forest loss add up to a meaningful conservation contribution.

Our findings demonstrate that protected areas have consistently reduced deforestation across the Amazon. By quantifying avoided forest loss at 1-km resolution, this study provides evidence directly relevant to conservation planning under the 30×30 framework. Beyond the Amazon, the approach can inform regional strategies, but its broader value will depend on tailoring models to local data and dynamics.

4 Materials and Methods

To characterize deforestation dynamics across the Amazon basin, we constructed a 1 km resolution annual deforestation dataset spanning 1986–2020, covering Brazil, Bolivia, Colombia, Ecuador, and Peru. Annual 30 m land cover maps from the MapBiomass Amazonia Project (MapBiomass, 2025) provided the foundation, from which we extracted forest and flooded-forest classes to produce binary forest-cover layers. We defined deforestation as the first-time forest-to-non-forest transition for each 30m pixel, retaining only primary deforestation events to focus on irreversible loss of original forest cover and excluding re-clearing of previously deforested areas. Binary deforestation maps were then aggregated to 1 km resolution, with cumulative deforestation constrained not to exceed each cell’s 1985 baseline forest cover. Full details of data construction and cleaning are provided in [Appendix B](#).

4.1 Covariates, Conservation Boundaries, and Sample Construction

Terrain variables (elevation, slope, aspect) were derived from the SRTM digital elevation model (Farr et al., 2007), distance to navigable rivers was computed from the Global River Database (Andreadis et al., 2013), and travel time to the nearest cities and ports from global accessibility datasets (cite), and all variables were standardized prior to model training. Potential agricultural revenue, calculated from attainable crop yields and annual global crop prices, and weighted with crop harvested area shares (cite), captures economic incentives for forest clearing. See [Appendix C](#) for more details of how these covariates are generated.

To capture sub-gridcell landscape structure, fragmentation, and land-use texture, we generated satellite image embeddings from a cloud-free Landsat 5 composite spanning 1985–1995 using DINOv3, a pretrained vision foundation model (Siméoni et al., 2025). These embeddings encode fine-scale spatial information (from 30 m Landsat imagery) that is lost at the 1 km resolution and complement the coarser-scale covariates described above. Full description of the embedding generation procedure are provided in [Appendix D](#).

Boundaries of Protected Areas (PAs) and other conservation designations, including Indigenous Peoples and Local Communities’ (IPLC) lands, private conservation areas (CONC), and Payment for Ecosystem Services (PES) zones were rasterized to the 30 m MapBiomass reference grid and aggregated to 1 km. A cell was classified as designated from the first year any portion of its area overlapped with a conserva-

tion boundary, with the earliest designation year assigned in cases of multiple overlapping designations. Full details of conservation boundary sources and processing are provided in [Appendix E](#).

To identify causal effects of protection, we constructed the analytical sample from two categories of cells: those that never received any conservation designation, and those eventually designated as PAs. Although PA cells and non-PA cells share the same broad Amazon context, they differ systematically in characteristics that predict both deforestation risk and the likelihood of receiving protection. To improve comparability, PA and non-PA cells were paired using propensity score matching on time-invariant and pre-treatment covariates, including historical forest cover, terrain, proximity to infrastructure, population density, and pre-protection potential agricultural revenue. For each PA cell, the matching procedure identified the two most similar non-PA cells, yielding a matched sample with substantially improved covariate balance (see [Appendix H](#)). Cells with IPLC, PES, or CONC designations were excluded from the control pool to avoid confounding effects from overlapping conservation policies. Each matched non-PA cell was assigned a pseudo-PA year by identifying the point in its deforestation history that most closely resembled the pre-designation trend of its matched PA cell over the preceding three years, providing a set of control gridcells whose counterfactual trajectories can be evaluated against known outcomes to validate the model. Our identification strategy rests on the assumption that, in the absence of protection, deforestation in PA cells would have developed in parallel with their matched controls, an assumption we validate by confirming pre-treatment trend alignment (see [Appendix H](#)).

4.2 Physics-constrained deep learning model for counterfactual prediction

Estimating what deforestation would have looked like in protected areas had they never received protection requires a model that can capture nonlinear, spatially heterogeneous deforestation dynamics across the full Amazon basin. We address this with a three-stage counterfactual prediction framework. Stages 1 and 2 are built around a physics-constrained, multimodal DL model that predicts deforestation occurrence and magnitude, while Stage 3 applies a random forest model to correct for systematic bias that accumulates when predictions are generated autoregressively over longer time horizons. We use the term physics-constrained to mean that predicted deforestation is bounded by the forest actually available; it can never be negative, it can never exceed the forest still standing in a gridcell, and any forest predicted to be lost in one year is subtracted from the forest available in the next. The DL model integrates four information streams: 10-year temporal sequences of deforestation and crop revenue processed by a shared Transformer encoder; time-invariant geographic covariates including terrain, accessibility, location, and population density processed through separate multilayer perceptrons; spatial neighborhood context captured from 12×12 km patches of historical mean forest cover and deforestation encoded by lightweight Vision Transformer (ViT) branches; and sub-gridcell landscape structure derived from satellite image embeddings, as described above.

The 12×12 km spatial neighborhood patches serve a role analogous to spatial lags in econometric models, but with greater expressiveness, since rather than collapsing neighborhood information into a single average, the ViT processes the full spatial arrangement of surrounding cells with positional encoding, enabling it to learn directional patterns such as deforestation fronts approaching from a particular direction. Together, each of the four feature streams produces a compact representation that is modulated by a country-level gating mechanism, which learns country-specific weights that adjust the relative importance of each feature stream before integration, allowing the model to capture qualitative differences in deforestation drivers across countries without imposing the constant additive shifts that country fixed effects in econometric models assume. Full architectural specifications and gating implementation details are provided in [Appendix J](#).

In Stage 1, a binary classifier estimates the occurrence of deforestation for each cell and year, generating a mask of likely deforestation events. Stage 2 then estimates deforestation magnitudes within masked cells using the same architecture, trained only on observations the classifier flagged as positive. A critical challenge in using DL for causal inference is that standard prediction objectives optimize for spatial accuracy but do not guarantee the properties required for valid counterfactual estimation. A model that minimizes prediction error may still produce systematically biased aggregate predictions or generate spurious trends that violate the parallel trends assumption underlying difference-in-differences designs. We therefore design a composite loss function in which each component addresses a specific requirement of our causal identification strategy:

$$L = \alpha L_{\text{WMSE}} + \beta L_{\text{corr}} + \gamma L_{\text{sum}} + \delta L_{\text{ZM}}.$$

The weighted MSE term (L_{WMSE}) provides the base spatial learning signal, but MSE alone is insuf-

efficient for valid counterfactual estimation. The correlation loss (L_{corr}) therefore counteracts regression to the mean, a common failure mode where models trained on MSE alone cluster predictions near the average and underpredict high-deforestation cells, by explicitly rewarding the model for preserving the relative spread and ranking of deforestation across gridcells. Building on this, the aggregate calibration term (L_{sum}) addresses the most critical requirement for causal validity, regularizing total predicted deforestation to match observed totals within each training batch and directly preventing systematic aggregate bias that would inflate or deflate estimated treatment effects. Finally, the zero-mass penalty (L_{ZM}) penalizes small positive predictions in cells where no deforestation occurred, training the model to predict exact zeros when covariates do not support forest loss rather than leaking small positive values that accumulate into a systematic upward bias across the many zero-deforestation cells at inference time. Full specifications of each loss component and final hyperparameter values are provided in [Appendix J.1](#). To enforce physical plausibility, the binary classifier receives available forest cover as an input feature, allowing it to learn that cells with no remaining forest do not deforest. The regression stage enforces this constraint directly in the architecture, bounding outputs so that predicted deforestation can never be negative or exceed the available forest cover. Together, these mechanisms ensure that model outputs respect basic ecological limits at every prediction step.

In Stage 3, a random forest model is applied as a bias correction step to address systematic errors that accumulate during autoregressive prediction. In autoregressive mode, each year’s predicted deforestation updates the available forest cover and deforestation in that year, which are used as input for the subsequent year. Small systematic errors compound over time under this design, causing the trajectory to drift from true values, an inference-time artifact that cannot be eliminated through improved training alone because it arises from the sequential feedback between predictions rather than from any single prediction step. To correct this, we train a random forest regressor on pseudo-PA gridcells whose true deforestation is known, learning the bias as a function of the model’s own prediction, the number of years elapsed since autoregressive prediction began, the calendar year, and the country. Corrected predictions are obtained by subtracting the predicted bias from cells the model flagged as deforestation, leaving predicted zeros unchanged, followed by a per-relative-year rescaling of the non-zero predictions that restores aggregate calibration. An equivalent bias correction is also estimated for the non-autoregressive predictions, which carry a smaller systematic bias of their own, so that both inference modes are calibrated consistently before comparison. Training and evaluation sets across all stages were defined using a spatial blocking strategy based on 100 km grids to reduce spatial autocorrelation. Full details of the DL model architectural specifications, training settings, and bias correction procedure are provided in [Appendix J](#), [Appendix K](#), and [Appendix M](#).

4.3 Average treatment impacts estimation based on predicted data

Despite using a matched dataset and a highly flexible DL model to predict counterfactual deforestation, year, country, and cell-specific characteristics might not perfectly be captured for every single location. Therefore, we apply econometric estimation techniques to control for these potential effects when assessing the impacts of PAs on deforestation. We need to account for the fact that PAs are established at different points in time, so our treatment is staggered. We address this using multiple approaches. First, we construct a variable that captures the relative time to treatment for each cell r . As long as we only look at cells that become PAs at some point, this assignment is complete for this data. We estimate the following model for the average treatment effect on the treated:

$$y_{itc} = \beta P_r + \gamma_{tc} + \epsilon_{itc}$$

where y_{itc} is the difference between observed deforestation and predicted deforestation, the residual, in cell i in calendar year t located in country c . Since we use these residuals directly as outcome variable and only cells that become protected are considered, every cell is in the treatment group. P_r is a binary variable indicating protection, and becomes 1 as soon as $r \geq 0$. β is the average treatment effect on the treated. γ_{tc} are year-by-country fixed effects and capture annual and country-specific developments, e.g. forest-related policies. ϵ_{itc} is the error term.

The DL model should capture the counterfactual deforestation in PA cells after protection well, given that it was also trained also on developments in PPAs at that time. Standard econometric approaches apply Difference-in-Differences (DiD) methods to control for developments in control observations after treatment. To assess the robustness of our average treatment estimate, we apply this DiD approach to the DL residuals using the following model:

$$y_{itc} = \alpha_i + \gamma_{tc} + \beta_1 T_i + \beta_2 P_r + \beta_3 (T_i \times P_r) + \epsilon_{itc}$$

where y_{itc} is the deforestation residual for grid cell i in year t located in country c from the DL model for PAs and PPAs. Cell fixed effects α_i account for location-specific deforestation characteristics. γ_{tc} are year-by-country fixed effects. T_i indicates membership of the treatment group, i.e. cells that eventually become protected. P_r indicates that the treatment is active, i.e. $r \geq 0$. For PPA cells this information is not naturally available. However, from our matching procedure we retrieve to which treatment years a grid cell was assigned. In case of multiple assignments as control unit, we select one hypothetical treatment year for each control cell, and then construct r for the PPAs. The interaction of these two variables $T_i \times P_r$ indicates 1, if a cell is in the treatment group and protection is implemented. β_3 captures the average treatment effect on the treated. ϵ_{itc} is the error term.

The estimations are carried out in R using the *fixest* package (Bergé, 2018). We use Conley standard errors at 100 km to account for spatial correlation. Despite the inclusion of γ_{tc} , the setup based on the relative time to treatment can cause biased estimates, as it does not consider the actual treatment timing explicitly and therefore already-treated cells may serve as inappropriate controls for later-treated cells (Callaway & Sant’Anna, 2021; de Chaisemartin & D’Haultfoeuille, 2020; Sun & Abraham, 2021). Therefore, we additionally run an estimation that accounts for these issues relying on the estimator suggested by Callaway and Sant’Anna (2021). We only use a random sample of the data for this computationally demanding estimation. We estimate DiD models for each PA cohort, comparing that cohort’s PA cells only to the never-treated PPA cells and dropping the year before PA establishment (see Figure Q1). The cohort-specific average treatment effects are then averaged with weights equal to each cohort’s share of treated cells, and the standard errors are calculated based on parametric bootstrap from a normal distribution and takes the SD of the resulting weighted averages.

4.4 Econometric model

As comparison to the counterfactual prediction approach, we use state-of-the-art econometric DiD estimations on the matched dataset. Such models have widely been used to evaluate protected area effectiveness (Martínez & Melo, 2025; Ribas et al., 2020; Silva et al., 2024). Identification of the average treatment effect on the treated is based on the assumption that, in absence of the treatment (i.e., protection status), deforestation would have developed parallel to what is observed in control areas also in then protected areas. To ensure comparability, we apply similar approaches as described in Section 4.3 for DiD estimations. Instead of using the DL model’s counterfactual predictions, here we directly use observed deforestation for PAs and PPAs as outcome variable, so that y_{it} is the deforestation outcome for grid cell i in year t located in country c . For these models, we add potential agricultural revenue R_{itc} as additional control variable. For the simple DiD approach, we exploit the information on relative time to treatment r to specify treatment exposure. Due to the staggered treatment time, we additionally estimate the staggered DiD model.

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Data, Materials, and Software Availability

All of the data documentation, data, and coding scripts necessary to replicate this project or reproduce the results of this paper have been deposited into this public GitHub repository:
<https://github.com/PratyushTripathy/amazon-deforestation-causal-ai>

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Appendix A Overall Workflow

This appendix provides a high-level overview of the data preparation and modeling workflow used in the study. Figure A1 summarizes the sequence of steps from raw data ingestion to the construction of model-ready inputs and evaluation outputs. The workflow emphasizes different steps of data matching and filtering, with exact number of data points at each step, which are central to the model training and counterfactual design of the analysis.

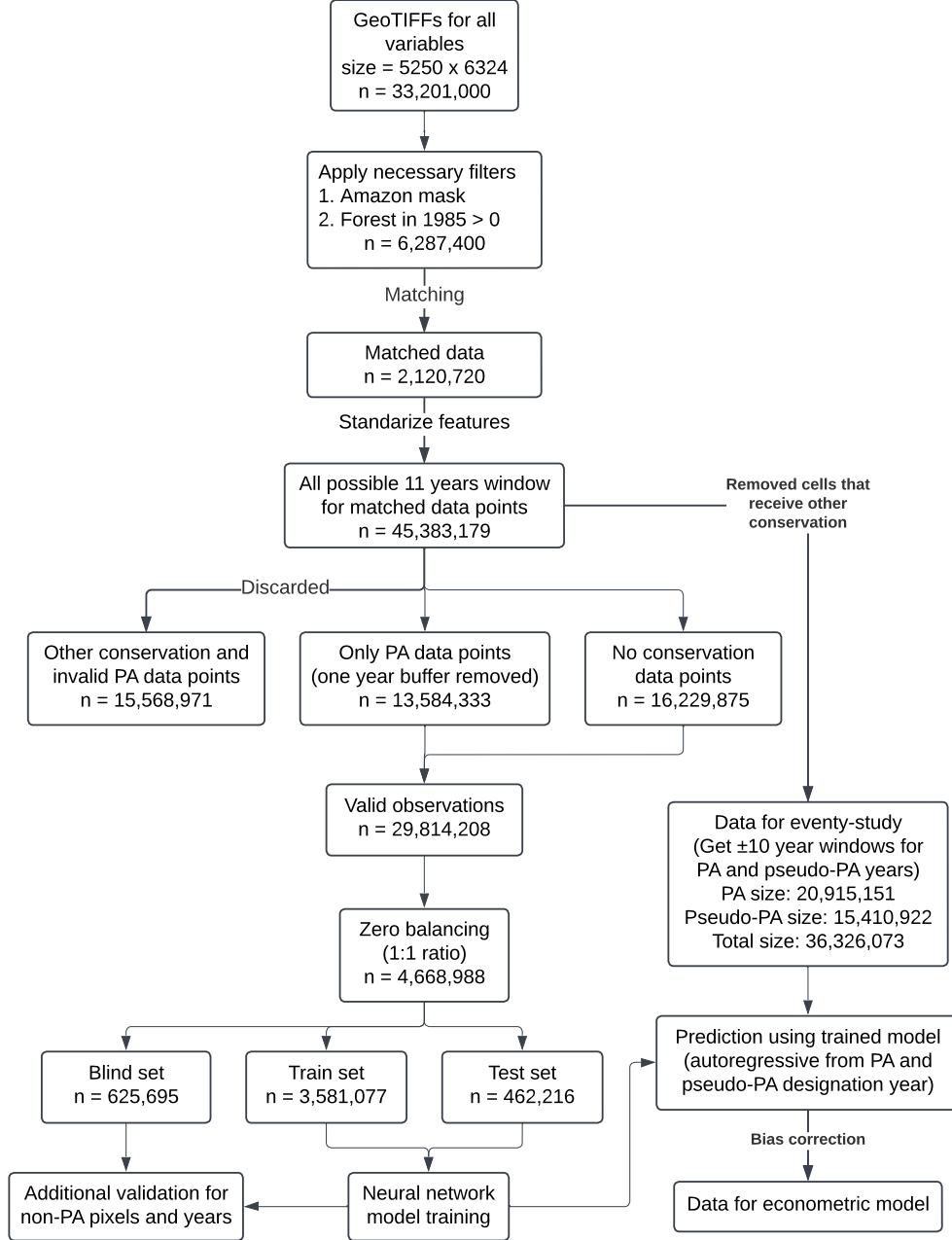


Figure A1: Data pre-processing workflow used in the present study.

Each major component of the workflow is described in detail in subsequent appendix sections. Data sources and preprocessing steps are documented in Appendix B, Appendix E, Appendix C and Appendix D, including the construction of deforestation and forest-cover time series, spatial masking, and quality control procedures. Model architecture and loss functions are provided in Appendix J and Appendix J.1, while training settings and accuracy assessment are described in Appendix K. The workflow diagram is intended to orient the reader and clarify how these components fit together, rather than to replace the

detailed descriptions that follow.

Appendix B Land-Cover and Deforestation Data

This section describes the construction of annual forest-cover and deforestation datasets used as primary inputs and targets in the analysis. Particular care is taken to ensure temporal consistency, physical plausibility, and a clear distinction between primary and secondary deforestation, which is central to our counterfactual evaluation framework.

Annual land-cover maps for 1985–2020 were obtained from the MapBiomass Project (Collection 6), which provides 30 m resolution land-cover classifications for Brazil, Bolivia, Colombia, Ecuador, and Peru (MapBiomass, 2025). Forest (class 3) and flooded forest (class 6) categories were merged to produce annual binary forest maps, where a value of 1 indicates forest presence and 0 indicates non-forest (Figure B1, panels a and b). Annual deforestation was computed by identifying pixels classified as forest in year t that transitioned to non-forest in year $t + 1$ (Figure B1, panel c).

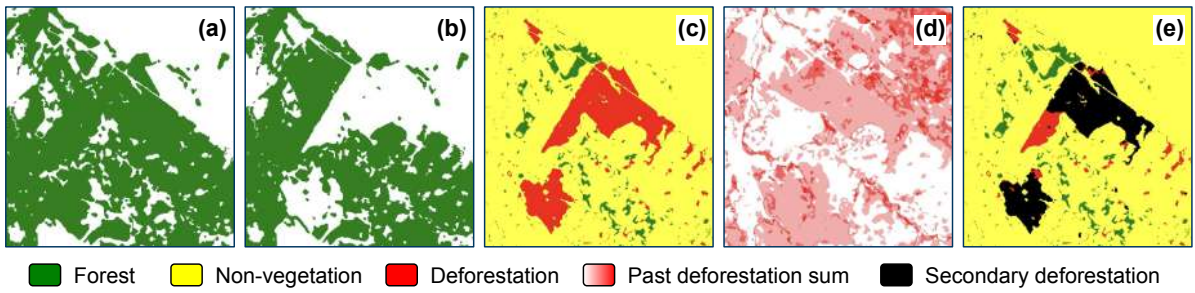


Figure B1: Illustration of deforestation calculation and classification. Panels show (a) binary forest cover for 2019, (b) binary forest cover for 2020, (c) deforestation identified in 2020, (d) cumulative deforestation prior to 2020, and (e) separation of primary and secondary deforestation for 2020.

Deforestation events were further classified as *primary* when corresponding to the first recorded forest-to-non-forest transition for a given pixel, and as *secondary* when occurring on land that had previously experienced deforestation and subsequent reforestation. This classification was based on pixel-level land-cover histories derived from the full time series (Figure B1, panels d and e).

To maintain analytical consistency and to focus on primary deforestation dynamics, we retain only the first deforestation event for each pixel and mask out any subsequent deforestation following reforestation. This filtering step ensures that all modeled deforestation represents irreversible loss of original forest cover within the study period.

Following this masking, the 30 m binary forest and deforestation maps were aggregated to a 1 km equal-area grid by computing the proportion of forested 30 m pixels within each cell (Figure B2). Given the 1 km² cell size, this proportion directly corresponds to forest area in square kilometers. This aggregation allows deforestation to be represented as a continuous quantity while preserving sub-pixel information.

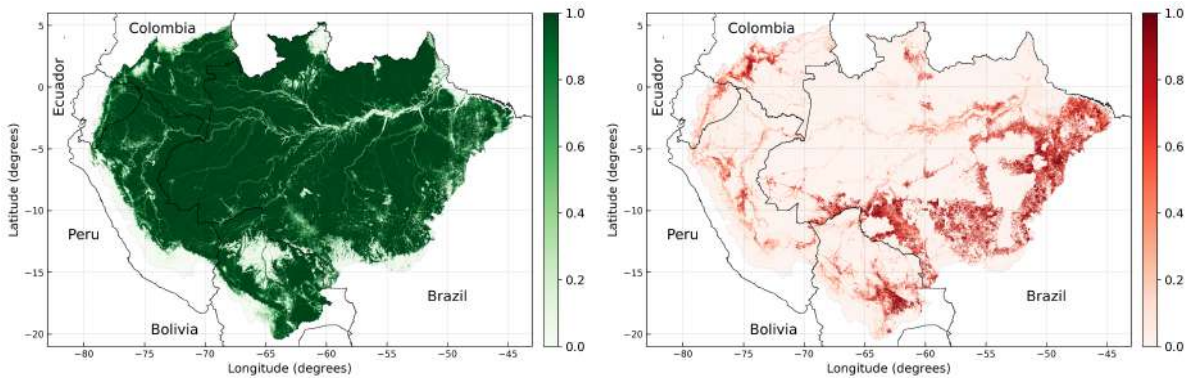


Figure B2: Maps showing (left) forest cover in 1985 and (right) total deforestation between 1985 and 2020 at 1 km resolution.

Although deforestation is derived from sequential annual forest-cover maps, reforestation is not incorporated symmetrically in this construction, which can lead to inconsistencies between forest-cover trajectories and cumulative deforestation.

To enforce temporal and physical consistency between forest cover and deforestation, we reconstruct forest dynamics using deforestation as the sole mechanism of forest loss. Specifically, we take forest cover in 1985 as the baseline and compute remaining forest in subsequent years by sequentially subtracting annual deforestation at the 1 km cell level. This sequential construction ensures that cumulative deforestation within any cell cannot exceed the forest area present at the beginning of the study period and that forest loss is represented consistently across both data products. Figure B3 illustrates forest cover and deforestation totals before and after cleaning, while Table B1 quantifies the resulting alignment between total forest loss and cumulative deforestation.

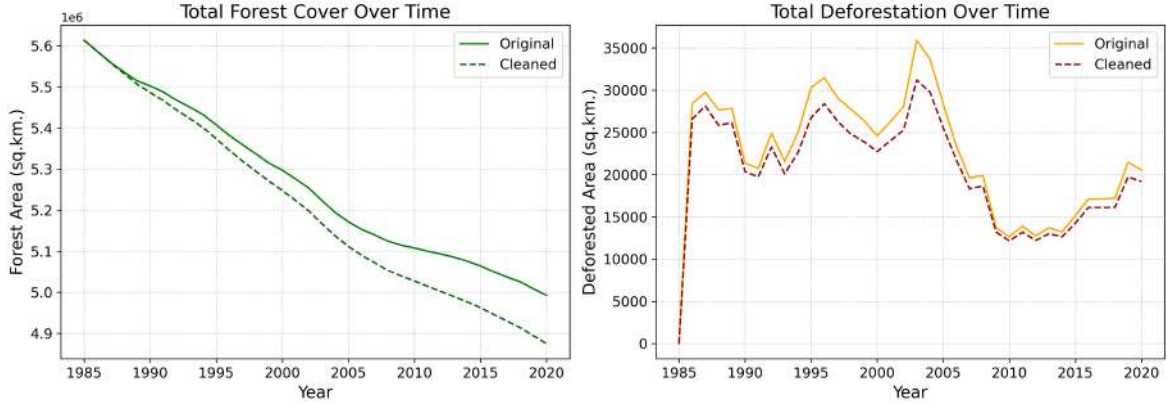


Figure B3: Annual forest cover and deforestation totals before (top) and after (bottom) cleaning.

Table B1: Comparison of total forest cover and deforestation (square kilometers) before and after cleaning.

	Initial Forest (1985)	Final Forest (2020)	Deforestation (1986–2020)	Forest Loss (1985–2020)	Residual Forest (Mismatch)
Original	5,613,279	4,992,857	800,592	620,422	-180,170
Cleaned	5,613,279	4,875,147	738,131	738,131	0

Appendix C Preparation of Environmental and Socioeconomic Covariates

To build a model capable of predicting deforestation, we integrate information beyond observed forest loss in order to capture the physical, economic, and accessibility-related factors that influence land-use change. Deforestation decisions are shaped not only by past forest dynamics, but also by terrain constraints, proximity to infrastructure and waterways, and economic incentives that affect the profitability of clearing forested land. We therefore construct a set of environmental and socioeconomic covariates designed to represent these drivers and provide additional context to the model.

To characterize variation in terrain across the study area, we derive elevation, slope, and aspect layers from the NASA Shuttle Radar Topography Mission (SRTM) Digital Elevation Model at 30 m resolution (Farr et al., 2007). These variables capture topographic conditions that can influence deforestation through their effects on land suitability, accessibility, and development costs.

Human accessibility is represented using proximity-based measures that reflect the ease with which forested areas can be reached and exploited. We generate distance-to-river layers using the Global River Database (Andreadis et al., 2013), and travel-time layers to the nearest cities and ports (Nelson et al., 2019). To ensure that river proximity reflects navigable waterways, we exclude rivers narrower than 50 meters, under the assumption that smaller channels are unlikely to support sustained boat or canoe traffic.

Economic incentives are captured in potential agricultural revenues, which reflects market-driven pressures that can increase the attractiveness of converting forested land to agricultural use. Potential agricultural revenue is calculated based on attainable crop yields in 2000 (Fischer et al., 2021) and annual global crop prices. Figure C1 provides an overview of the resulting spatial inputs, including terrain, accessibility, and economic variables used in the analysis.

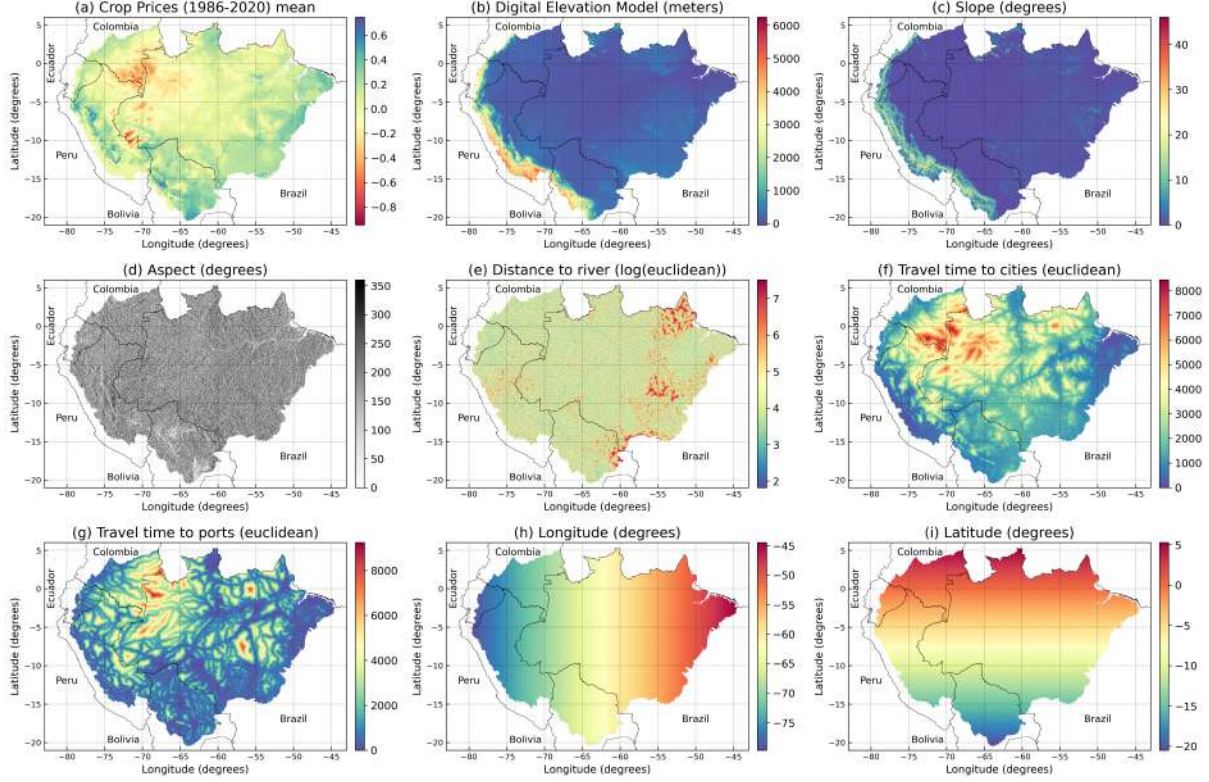


Figure C1: Maps showing (a) crop prices averaged over 1986–2020, (b) digital elevation model, (c) slope, (d) aspect derived from the DEM, (e) log distance to rivers, (f) travel time to the nearest cities, and (g) travel time to the nearest ports.

Appendix D Generating satellite embeddings using DINOv3

To generate a cloud-free baseline composite of historical imagery, we constructed a pixel-based 25th percentile composite from Landsat 5 Collection 2 Level-2 surface reflectance data over the Amazon basin. All Landsat 5 images acquired between 1985 and 1995 with less than 30% reported cloud cover were retrieved. Following Corbane et al. (2020), who developed a global cloud-free composite from Sentinel-2 by applying stringent cloud thresholds and compositing rules, we applied the pixel quality assessment (QA_PIXEL) band to mask clouds, shadows, and snow. The masked image collection was reduced using the 25th percentile per pixel and per band. This percentile statistic provides a robust estimate of the least cloudy observations while mitigating the influence of outliers from residual haze or shadows. Figure D1 below shows the resulting composite in true color, with the 25th percentile values representing decadal surface reflectance conditions across the Amazon basin.



Figure D1: Landsat 5 decadal composite (1985–1995) displayed in true color. Each pixel shows the 25th percentile of cloud-free surface reflectance values, providing a robust representation of long-term conditions across the Amazon basin. Generated using <https://code.earthengine.google.com/f5938937bf28b0958bcbeaa5eb99d64c>

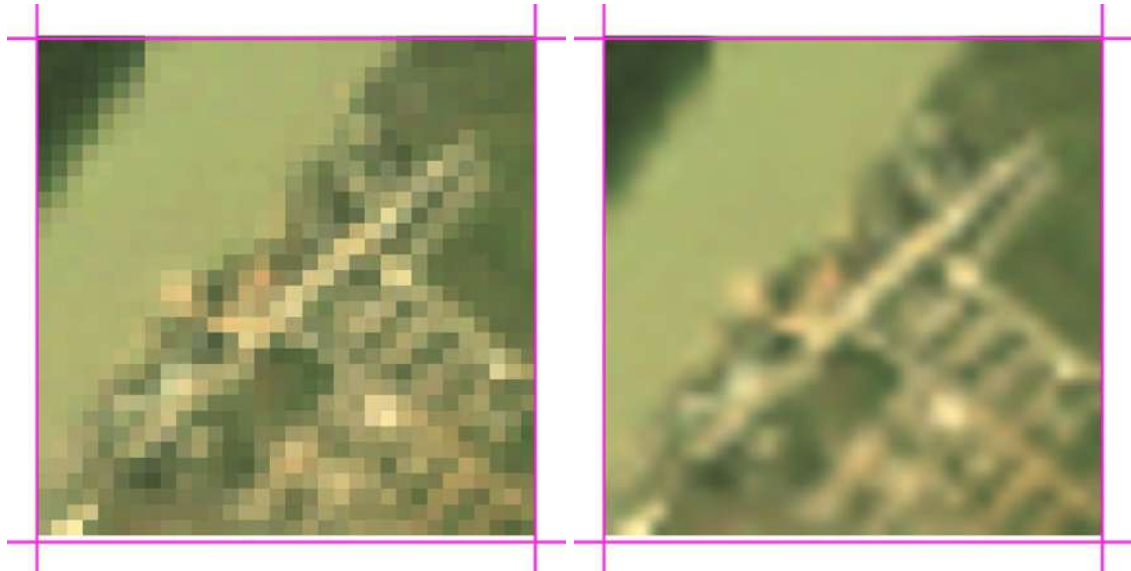


Figure D2: Figure showing (left) the original 32x32 Landsat chip at 31.25 m resolution for a 1 km grid and (right) a resampled version of size 512x512 pixels generated using bicubic resampling. The DINOv3 backend uses an image size of 16x16 so a square image chip of size 32 would produce only 4 tokens, which wouldn't be enough to extract meaningful embeddings. Using a square chip of size 512 gives 1024 tokens to extract embeddings from.

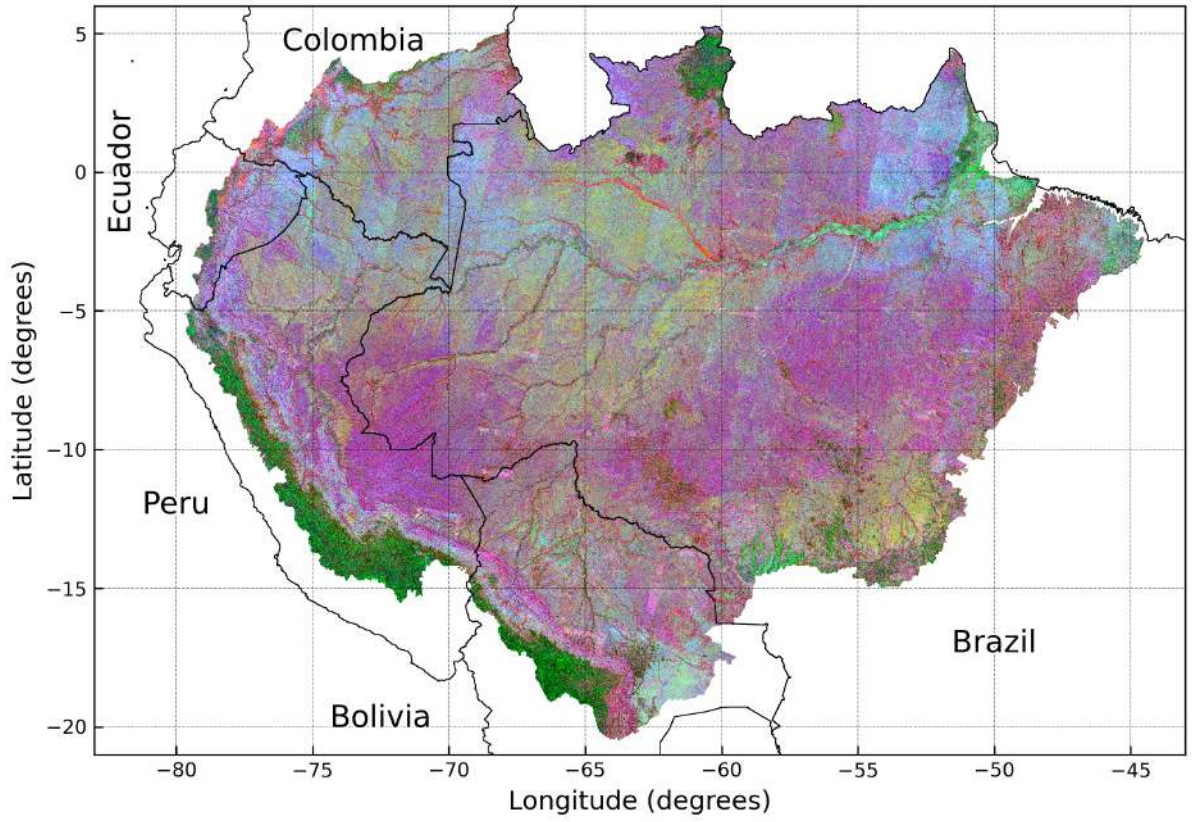


Figure D3: DINOv3-derived satellite embeddings for the Amazon region. The embeddings contain 1024 features; the map shows features at indices 52, 65, and 421 as an RGB composite.

Appendix E Protected Areas and Other Conservation Designations

Protected Areas (PAs) constitute the primary policy intervention evaluated in this study. In addition, we compile spatial data on other conservation designations, including Indigenous Peoples and Local Communities' (IPLC) lands, Payment for Ecosystem Services (PES) areas, and Conservation Concessions (CONC). While these designations are not the focus of the analysis, they play an important methodological role by allowing us to construct control groups that are not affected by overlapping conservation policies. Throughout the analysis, grid cells influenced by non-PA conservation designations are excluded from the control pool to isolate the effects of Protected Areas (see [Appendix H](#)).

All conservation boundaries were obtained from national and regional sources, summarized in [Table E1](#), projected to a common coordinate reference system, and rasterized to 30 m resolution to align with the land-cover data.

Table E1: Sources of conservation designation datasets used in the study.

Designation Type	Spatial Coverage	Temporal Coverage	Data Source
PA	Amazon Basin	1985–2020	[To be filled]
IPLC Lands	Amazon Basin	1985–2020	[To be filled]
PES Areas	Amazon Basin	1985–2020	[To be filled]
CONC	Amazon Basin	1985–2020	[To be filled]

Annual conservation layers were produced for each year from 1985–2020 by overlaying the rasterized boundaries. For each 1 km grid cell, we computed the proportion of area covered by each conservation designation in each year. A cell was classified as belonging to a given designation if any portion of the cell overlapped with that designation, and the first year in which this condition was met was recorded as the designation start year. In cases where multiple conservation designations overlapped within a single cell, we assigned the earliest designation year and labeled the cell according to the designation with the highest proportional coverage in that year. This procedure yields a consistent temporal definition of treatment while ensuring that cells affected by multiple conservation policies are handled systematically.

[Figure E1](#) shows the spatial distribution of conservation designations across the study region. Protected Areas are displayed prominently to reflect their central role in the analysis, while IPLC lands, PES areas, and Conservation Concessions are shown for reference. These ancillary designations are used to identify and exclude potentially contaminated control cells rather than to estimate treatment effects directly.

In addition to spatial variation, Protected Areas exhibit substantial heterogeneity in the timing of designation across countries. [Figure E2](#) summarizes the distribution of PA implementation years by country, highlighting the staggered rollout of protection across the Amazon basin.

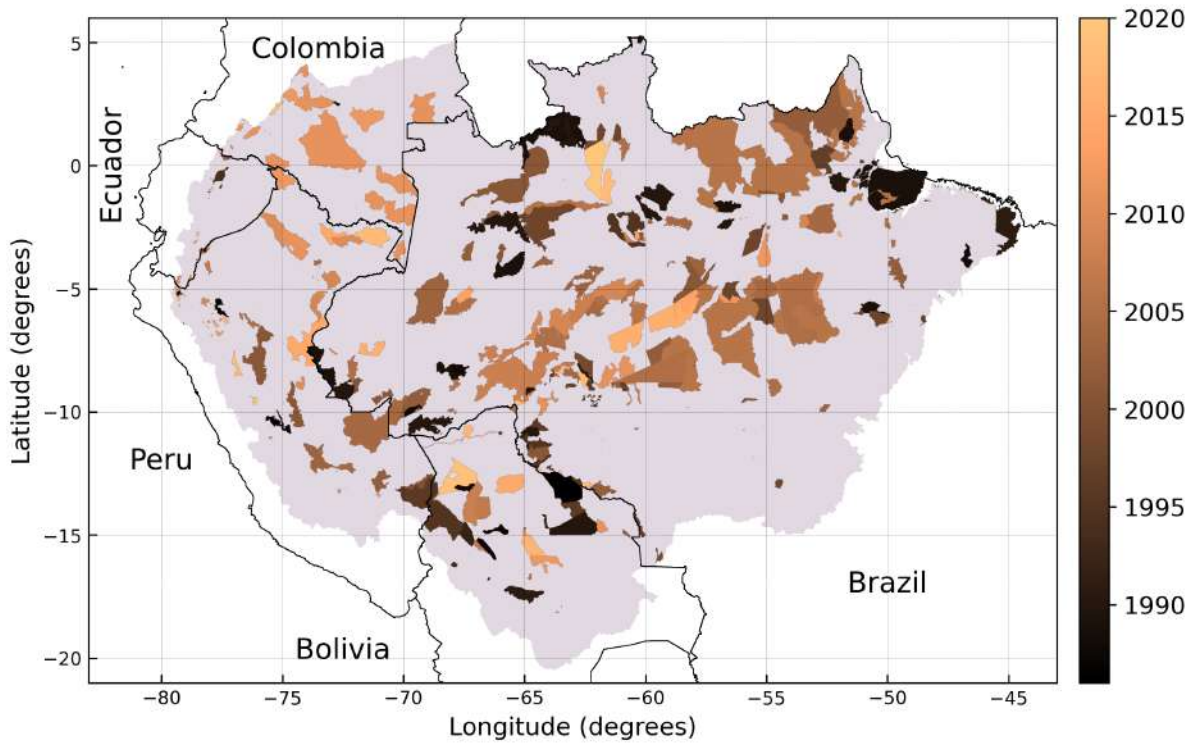


Figure E1: Spatial distribution of conservation designations in the Amazon. Panel (a) shows the first year of Protected Area designation. Panels (b–d) show the first year of designation for Indigenous Peoples and Local Communities' lands (IPLC), Payment for Ecosystem Services (PES) areas, and Conservation Concessions (CONC), respectively.

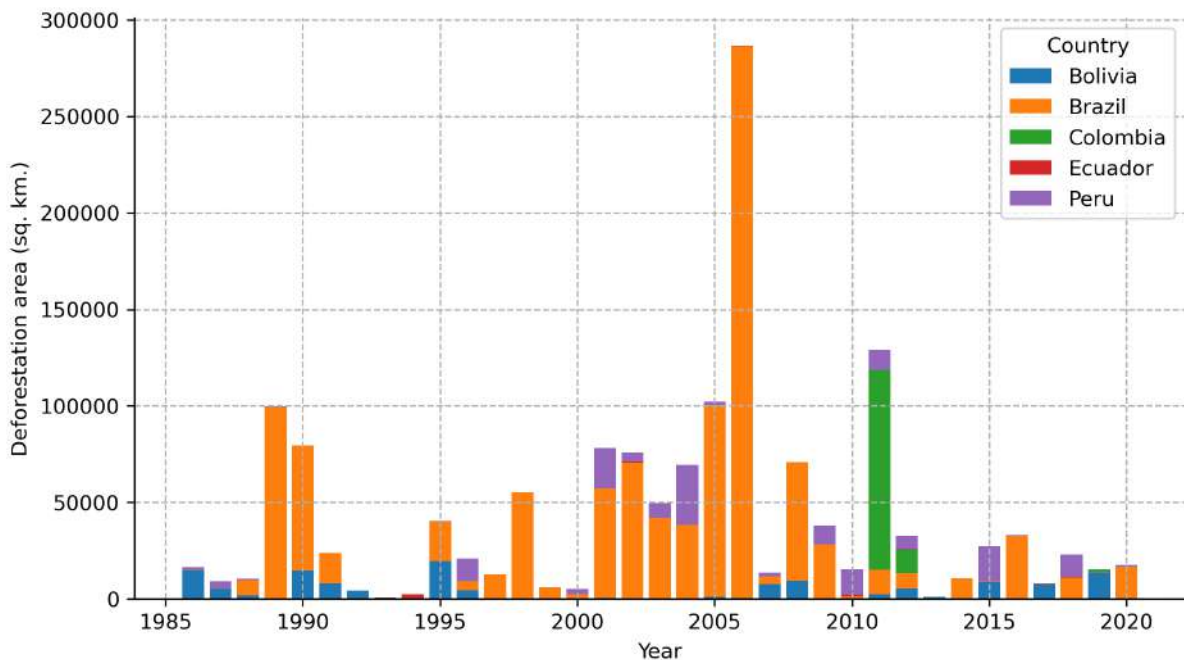


Figure E2: Distribution of Protected Area implementation years by country. Bars indicate the number of grid cells whose first year of PA designation falls in a given year.

Appendix F Spatial split for cross-validation

To keep the data for training the DL model different from an independent validation dataset used during every epoch, and to have another independent testing set that the model has not seen, we split our data into training-validation-test sets. To have a blind set of data kept only for the assessment of the final model, we create a blinded set. Therefore, we split our entire data into training (70%), validation (10%), testing (10%) and blind (10%) sets. While our data is at 1 km resolution, to avoid bias in assessment due to spatial autocorrelation, we split our data using 100 km resolution grids (Figure F1).

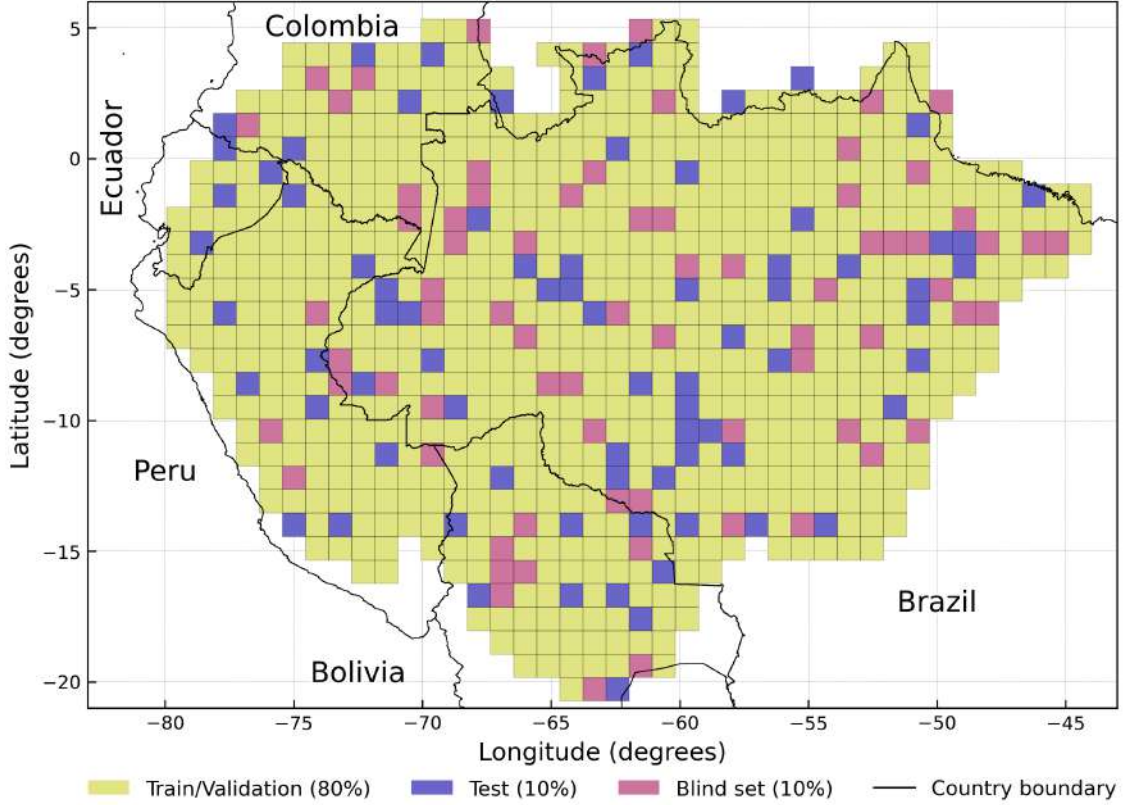


Figure F1: Spatial split of the Amazon into training/validation, test, and blind held-out sets using 100 km grid blocks. Blocks are assigned roughly 80% to training and validation combined, 10% to test, and 10% to a blind held-out set. Within the training and validation portion the pixels are further split into about 70% training and 10% validation of the total.

Appendix G Defining Treatment and Control Groups

To train a model capable of predicting deforestation without contamination from post-treatment information, we constructed the dataset from two categories of grid cells. The first category consisted of cells that never received any conservation designation; observations from these cells were included for the entire period 1986–2020. The second category consisted of cells that were eventually designated as PAs, but only observations ending at least one year before the year of designation were included. This one-year buffer prevented the model from being influenced by pre-announcement behavioral changes.

For both categories, observations were structured into rolling 11-year temporal windows. Each window comprised the first 10 years as predictor variables and the 11th year as the prediction target (Figure G1). This rolling design generated multiple overlapping sequences for each cell, allowing the model to learn from a variety of temporal contexts. For example, a PA cell designated in 2000 would contribute windows ending in 1999, 1998, and earlier, but no sequences extending into or beyond 2000. This ensured that every PA sequence represented strictly pre-treatment conditions and excluded any influence from conservation implementation or anticipation effects. In contrast, non-PA cells contributed windows drawn across the entire 1986–2020 period, providing coverage of both low- and high-deforestation eras.

726 This combination exposed the model to a wide range of baseline deforestation patterns while preserving
727 causal validity for treated cells.

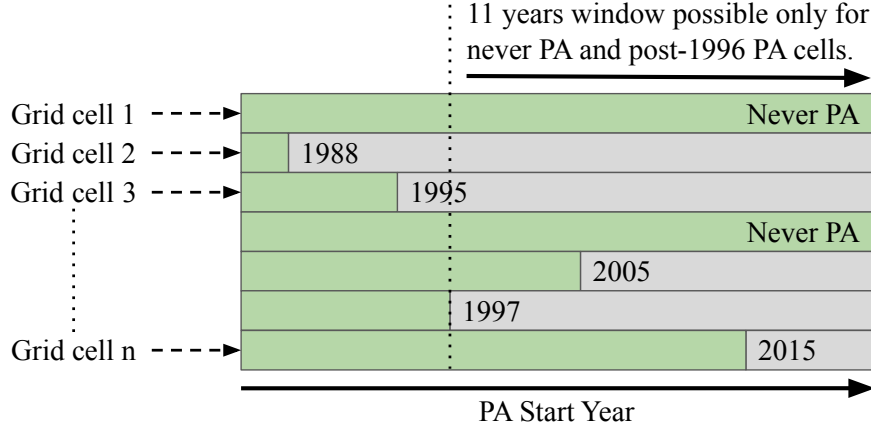


Figure G1: Schematic illustration of temporal window creation. Grid cells are shown with varying PA start years and never-PA status. Green indicates periods used for predictive modeling, while gray denotes periods excluded due to PA designation or buffer years. Only observation windows occurring entirely before PA designation are included for ever-PA cells.

728 Appendix H Data Filtering and Matching

729 We apply a sequential data filtering and matching procedure to define the analytical sample used in
730 subsequent analyses. Starting from the full set of 6,685,159 pixels, we first remove all pixels with zero
731 forest cover in 1985, leaving 6,287,400 pixels (Figure H1). We then apply propensity-score nearest-
732 neighbor matching, yielding 2,120,720 pixels. Finally, we exclude all pixels assigned to the blind set,
733 leaving 1,879,402 pixels used for model training and evaluation.

734 The matching step above improves comparability between treatment and control observations. We
735 apply propensity score-based nearest neighbor matching using time-invariant and pre-treatment charac-
736 teristics, including historical forest cover, historical deforestation levels, potential agricultural revenue
737 prior to protection, slope, elevation, and distances to major cities, ports, and rivers. Cells with Indige-
738 nous Peoples and Local Communities (IPLC), Payments for Ecosystem Services (PES), or conservation
739 concession (CONC) designations are excluded once they retrieve this land status to avoid confounding
740 institutional effects. The matched sample is constructed such that, for each treated cell there exist two
741 control cells. The matching is carried out based on the *MatchIt* package (Ho et al., 2011) in *R* and for
742 each i) treatment year and ii) local administrative zone independently. Therefore, control cells can be
743 matched to various treated cells. This procedure yields a set of protected-area (PA) and non-PA pixels
744 with comparable pre-treatment characteristics, improving covariate overlap and sample balance.

745 Figure H2 shows that average deforestation in treated cells and their matched controls is considerably
746 more similar than in unmatched cells, particularly in earlier years of the study period. Observations in
747 the treated group become fewer over time as already-protected cells are excluded from this visualization.
748 Furthermore, the covariate distribution of treated and matched control cells is more similar to each
749 other than to the respective distributions of unmatched cells (Figure H3), confirming that the matching
750 procedure yields a well-balanced analytical sample.

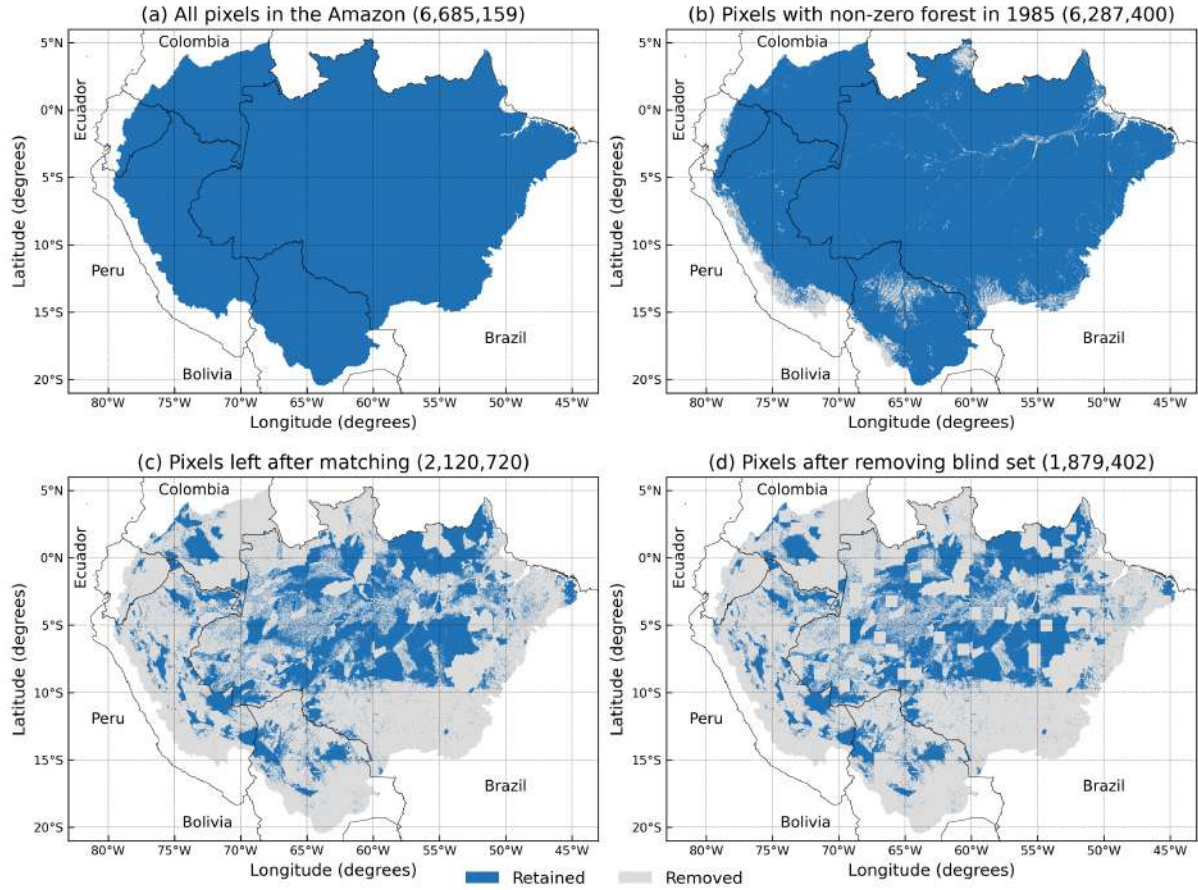


Figure H1: Spatial distribution of pixels at each stage of data filtering and matching. Panels show (a) all pixels in the Amazon, (b) pixels with non-zero forest cover in 1985, (c) pixels remaining after matching, and (d) pixels remaining after also removing the blind held-out set. Retained pixels are shown in blue and removed pixels in grey. Parentheses contain the number of pixels retained at each step.

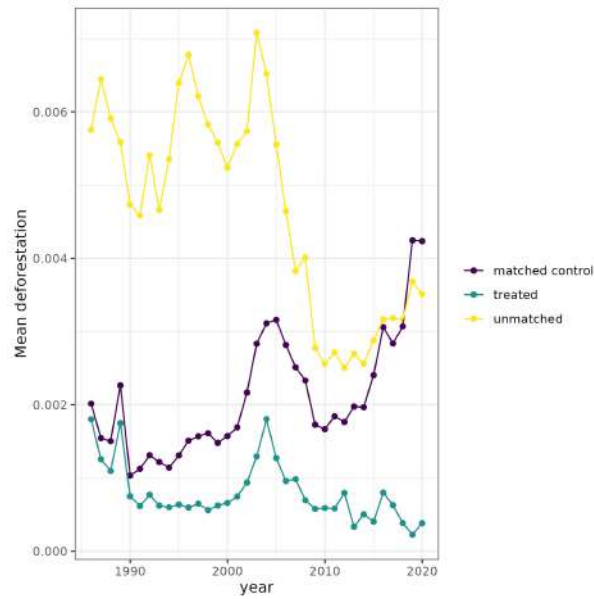


Figure H2: Mean deforestation by year in treated cells, matched control cells and unmatched cells. Treated cells are only captured here before they became protected.

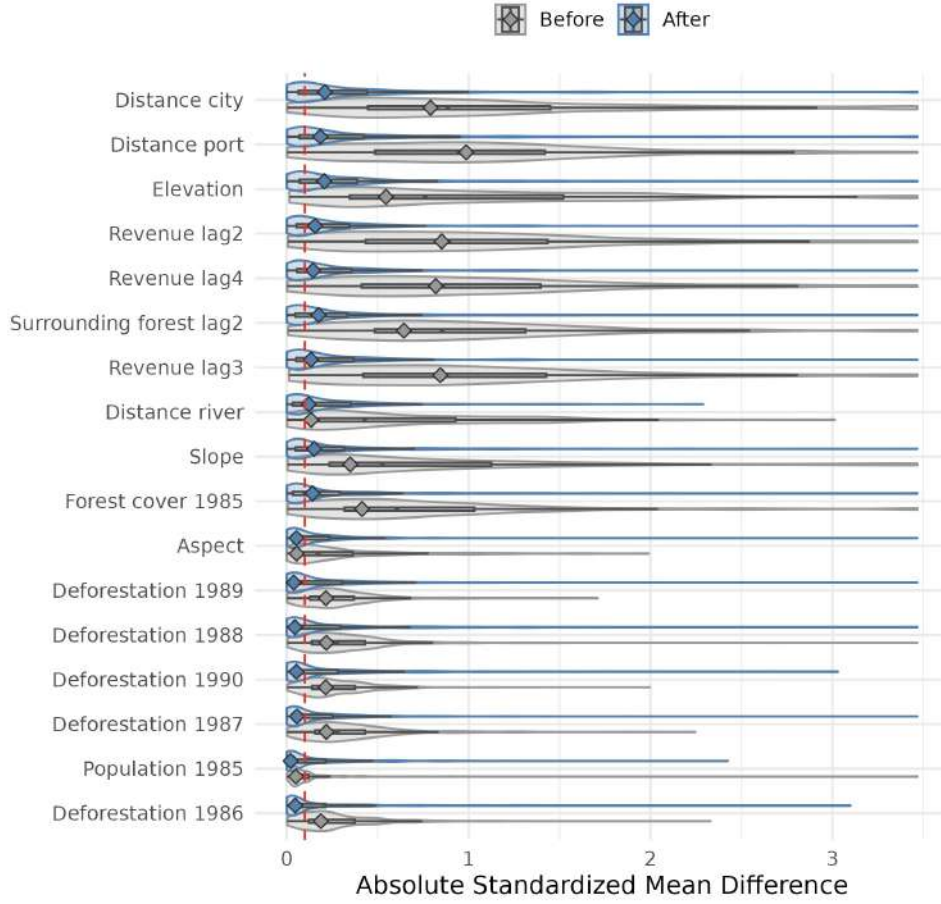


Figure H3: Matching balance shows reduced difference in covariate levels between treated cells and controls after matching

Appendix I Data Balancing

Generating all possible 11-year rolling-window for matched cells ($n=1,248,139$) produced about 40 million cell-year observations. The target variable was highly imbalanced, with approximately 93% of observations (37 million) having zero deforestation in the target year. To address the class imbalance arising from an excessive proportion of zero values (93%) in the original training dataset, we performed data balancing by randomly downsampling the zero-valued observations. Specifically, zero-valued data points were randomly selected to achieve a 1:1 ratio with the non-zero observations, thereby reducing the proportion of zero values from 93% (37,245,873 out of 40,069,729 total observations) to 50% (1,990,518 zero-valued observations). This resulted in an overall reduction of the dataset from approximately 40 million observations to 4 million observations. To ensure that the data balancing step (random sampling) does not create imbalance of observations across time, we plot Figure I1 that shows the temporal distribution of observations before and after data balancing. As Figure I1 shows, the ratio of largest and smallest number of observation is about 1:2, which is a good balance in data temporally.

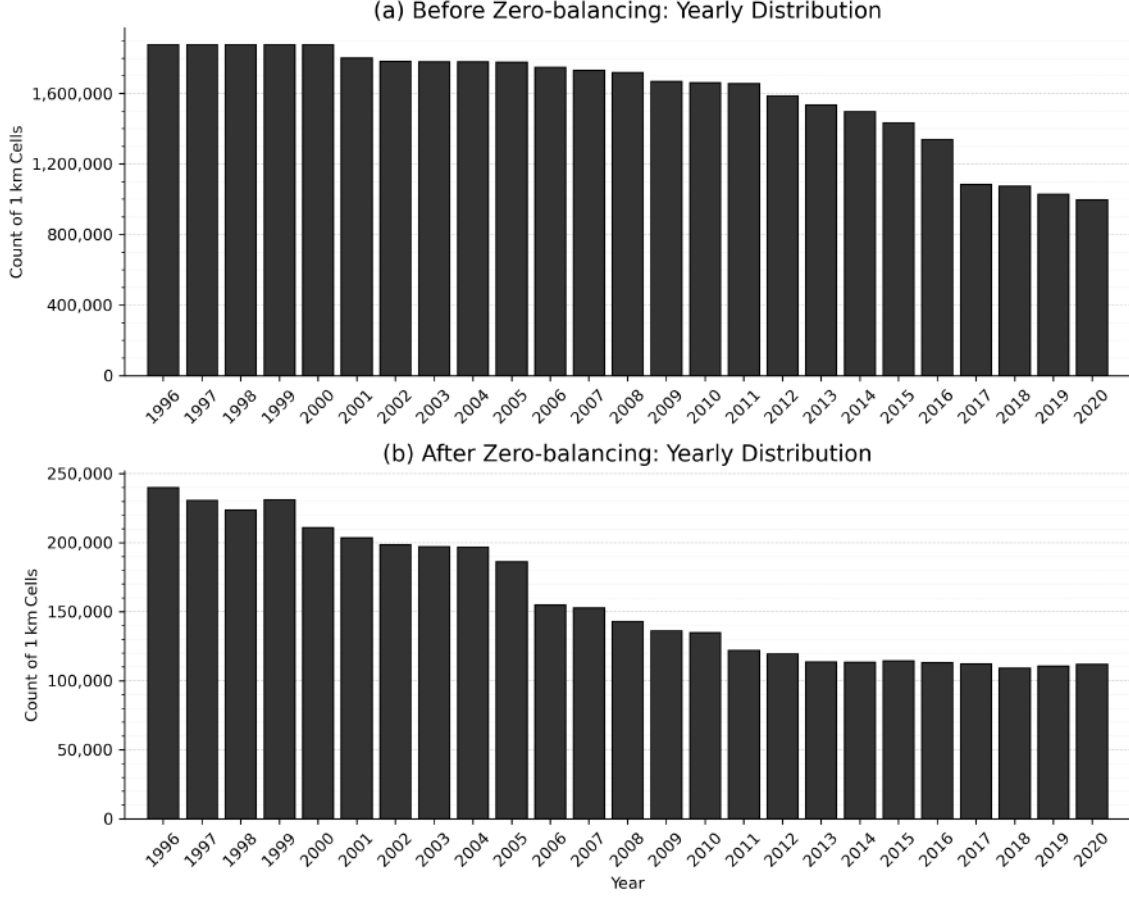


Figure I1: Temporal distribution of observations before (top) and after (bottom) zero-balancing the outcome variable by year.

Appendix J Deep learning Model Architecture

The modeling framework is built around a two-stage prediction strategy designed to separately estimate the occurrence and the magnitude of deforestation. Both stages share the same core architecture and input representations, but are implemented as two distinct models applied sequentially. The first model predicts whether deforestation occurs in a given grid cell and year, while the second model predicts the amount of deforestation, conditional on a positive prediction from the first stage. This design reflects the rarity and zero-inflated nature of deforestation outcomes and provides a clear separation between detection and magnitude estimation.

Formally, predicted deforestation for grid cell i in year t is given by

$$\hat{y}_{i,t} = \hat{z}_{i,t} \cdot f_{\theta}(x_{i,t}), \quad \hat{z}_{i,t} = \mathbf{1}(p_{\phi}(x_{i,t}) > \tau),$$

where $p_{\phi}(\cdot)$ denotes the binary classifier, $f_{\theta}(\cdot)$ denotes the regression model, and τ is a fixed probability threshold, which defaults to 0.5.

At the core of both stages is a physics-constrained hybrid neural network that integrates temporal, spatial, and static information through dedicated processing branches. Features from each branch are combined through a gated fusion mechanism before entering the fully connected layers that produce each model's single output. The two stages are separate instances of this identical architecture, carrying 153,886 trainable parameters each and sharing no weights, and they differ only in their output activation and in the physical bound applied to the regression. Figure J1 summarizes the architecture.

Temporal dynamics are modeled using Transformer branches that process 10-year sequences of historical deforestation and crop revenue. Both series pass through the same encoder. The positional encoding, the input layer normalization, and the Transformer encoder itself are shared parameters, and only the input projection and the final fully connected layer are learned separately for each series. The positional encoding is a fixed sinusoidal function and it is applied before the input layer normalization.

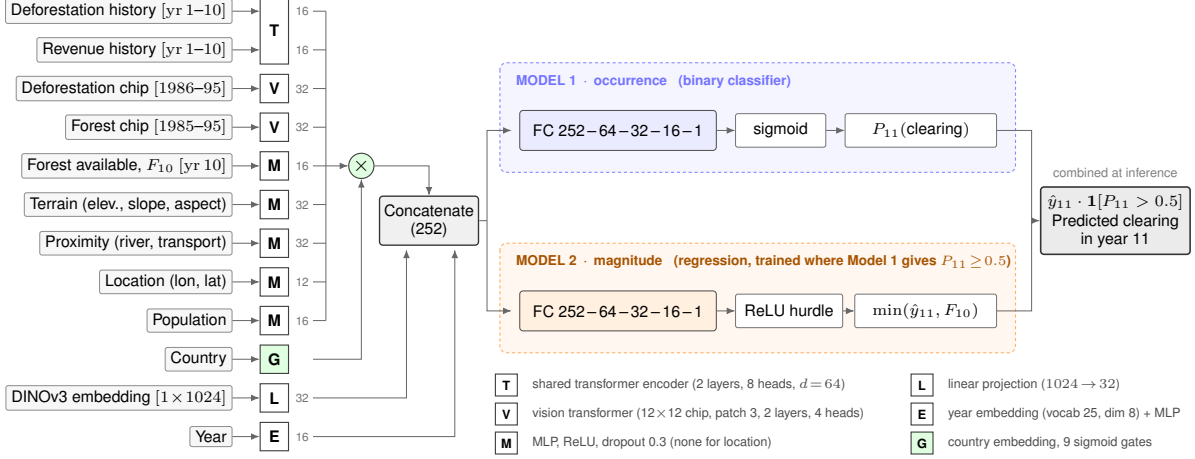


Figure J1: Architecture of the physics-constrained deep learning model.

The encoder output is mean-pooled across the time dimension and projected to a 16-dimensional feature vector with ReLU activation. Deforestation targets and forest availability values are scaled by a factor of 100 during training to improve numerical conditioning.

Time-invariant geographic attributes are modeled using five static feature branches, each designed to capture a distinct aspect of spatial heterogeneity. Terrain variables (elevation, slope, and aspect), proximity measures (distance to major rivers and two transport infrastructure distances), geographic location (latitude and longitude), population density, and forest availability in the final year of the input window, denoted F_{10} , are processed through separate multilayer perceptrons with ReLU activation and dropout. The location branch is the one exception, consisting of a single linear layer with ReLU and no dropout. This design allows the model to learn distinct representations for different classes of static covariates before they are integrated through the feature fusion mechanism. All static features are standardized prior to model training.

To provide spatial neighborhood context around each target grid cell, we supply the model with local image chips that summarize surrounding forest structure and disturbance patterns. Specifically, we construct two inputs at 1 km resolution: a forest-cover chip and a deforestation chip, each formed by taking temporal averages over an early reference period. Forest chips represent mean forest cover over 1985–1995, while deforestation chips represent mean annual deforestation over 1986–1995. These 12×12 pixel chips are encoded using two parallel Vision Transformer branches with independent weights, one for forest cover and one for deforestation. Each branch applies a convolutional patch embedding and adds learnable positional embeddings. The encoder carries no class token, so its layer-normalized output is mean-pooled over the patch tokens to a 32-dimensional feature vector. Using historical temporal averages captures stable spatial patterns while avoiding information leakage from future observations.

Additional contextual information is incorporated through separate embedding branches that are integrated at the feature fusion stage. Calendar year is represented using a learned embedding followed by a two-layer MLP. DINOv3 embeddings (1024-dimensional feature vectors extracted from a self-supervised vision foundation model, (Siméoni et al., 2025)) are projected into a 32-dimensional space through a two-layer MLP with ReLU activation and dropout, allowing the model to leverage rich spatial information learned from remote sensing data.

To account for regional heterogeneity in deforestation processes, we introduce an administrative-region gating mechanism. Each of the five countries in the study area is represented by a learned embedding, which is passed through a two-layer gating network that produces nine scalar gates in $[0, 1]$. These gates modulate the contribution of nine feature blocks (deforestation history, crop revenue, forest availability, terrain, proximity, location, forest ViT, deforestation ViT, and population) during feature fusion. Every branch output is layer-normalized before fusion, so that the gates operate on comparable scales. Year features and DINOv3 embeddings bypass the gating mechanism and are concatenated directly. This design allows the model to adapt the relative importance of different information sources across countries while retaining a shared global architecture.

Both stages produce a single output per grid cell and year. The first-stage model outputs the logit of deforestation occurrence, which is passed through a sigmoid at inference to yield a probability. The second-stage regression model applies a ReLU hurdle, transforming the raw output m as $\hat{y} = \max(0, m - \epsilon)$

with a fixed $\epsilon = 0.001$, which encourages the model to produce exact zeros for non-deforestation cells. The regression output is then clamped to F_{10} , so that predicted deforestation cannot exceed the forest remaining in the cell. These constraints keep predictions physically plausible and consistent with observed forest dynamics.

Appendix J.1 Loss functions for the DL model

Binary classification loss (Stage 1) The first-stage binary classifier is trained using binary cross-entropy loss with logits (BCEWithLogitsLoss), with a positive-class weight of 2.0 to account for the imbalanced prevalence of deforestation events. This weighting doubles the gradient contribution from true deforestation cells, encouraging higher recall at a modest cost to precision.

Regression loss (Stage 2) The second-stage regression loss is described in the main text. The full specification with default hyperparameters is:

$$\mathcal{L} = \alpha \mathcal{L}_{\text{WMSE}} + \beta \mathcal{L}_{\text{corr}} + \gamma \mathcal{L}_{\text{sum}} + \delta \mathcal{L}_{\text{ZM}}$$

with $\alpha = 1.0$, $\beta = 0.2$, $\gamma = 15.0$, and $\delta = 2.0$. The weighted MSE applies importance weights $w_j = (y_j + 1)^{0.25}$ (clamped to be non-negative), emphasizing cells with higher deforestation magnitudes without allowing extreme outliers to dominate the gradient. The masked correlation loss is computed only over observations where $y_j > 0$, ensuring that the spatial pattern signal is not diluted by the large number of zero-valued cells. The aggregate calibration term is $\mathcal{L}_{\text{sum}} = \left(\frac{\sum_j \hat{y}_j - \sum_j y_j}{\sum_j y_j + \epsilon} \right)^2$, the squared relative error of the batch totals, and the zero-mass penalty is $\mathcal{L}_{\text{ZM}} = \frac{1}{|Z|} \sum_{j \in Z} \hat{y}_j^2$ over the set of zero-deforestation cells $Z = \{j : y_j = 0\}$.

Appendix K Deep learning model training and performance

Both stages were trained using the Adam optimizer with an initial learning rate of 10^{-4} , weight decay of 10^{-4} , and mixed-precision training (FP16) for computational efficiency. The learning rate was scheduled using cosine annealing (CosineAnnealingLR), decaying smoothly from the initial value to $\eta_{\min} = 10^{-6}$ over the full training run. Mini-batch sizes were set to 5120 for both stages. The training set was split 80/20 into training and validation sets. Models were trained for up to 100 epochs with early stopping if validation loss did not improve for 25 consecutive epochs. The full balanced training dataset was used, with no further downsampling at the training stage. Both stages were trained sequentially in a single script: the binary model is trained first, then its predictions are used to mask the training data for the regression stage (only cells predicted as positive are included). All experiments were conducted on an NVIDIA A40 GPU with 48 GB memory.

Model performance was evaluated using accuracy, precision, recall, F1 score, and ROC-AUC for Stage 1, and root mean squared error, mean absolute error, R^2 , Pearson correlation, and aggregate calibration (predicted-to-actual deforestation sum ratio) for Stage 2. Feature block contributions were inspected by analyzing learned gate weights across administrative regions.

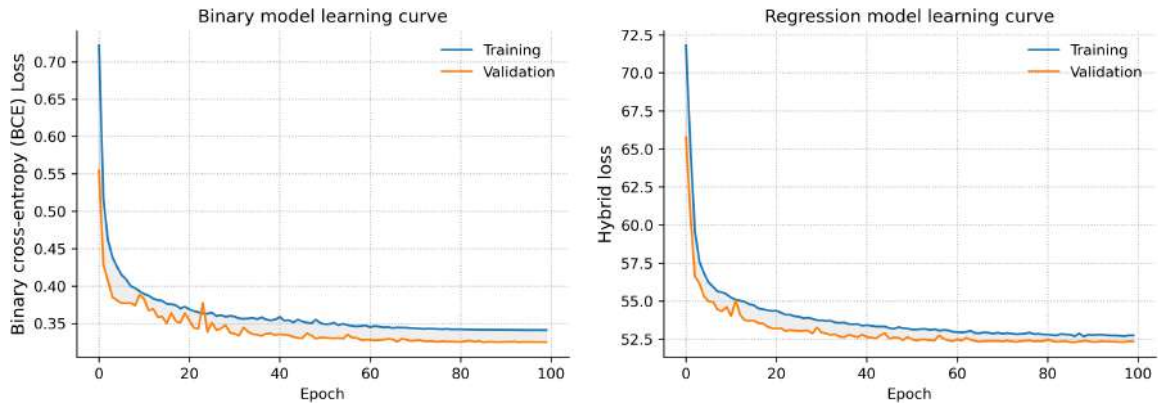


Figure K1: Learning curve of the binary and regression models.

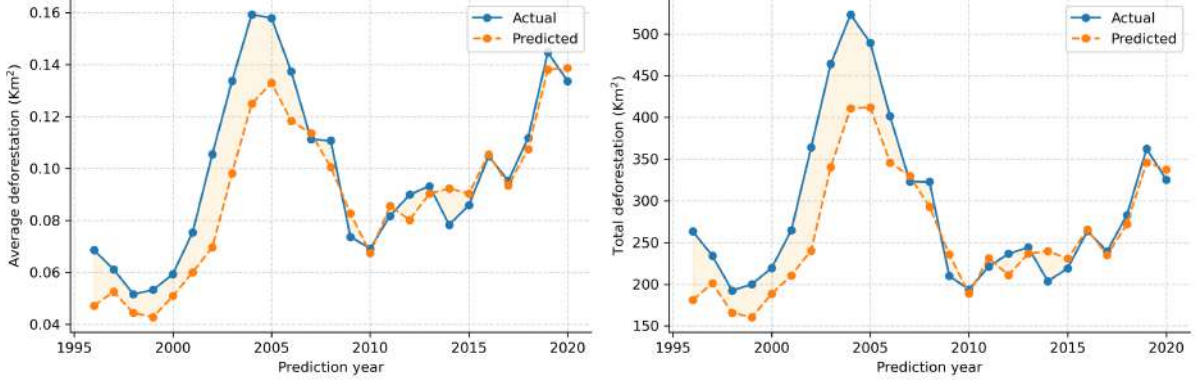


Figure K2: The line plot compares observed and predicted annual deforestation per grid cell, illustrating close agreement and consistent tracking of interannual variability.

Appendix L Out-of-sample comparison with the econometric model

To place the deep learning model’s performance in context, we apply the same metrics to predictions from an econometric model on deforestation. These metrics are meant to provide a reference point for predictive accuracy, not an assessment of the predictive capabilities of this linear and naïve econometric model. To be able to predict out-of-sample, we cannot use the econometric DiD model that we used for assessing the average treatment effect for the following reasons. First, the econometric model was estimated on a random sample of the matched data, not based on an exclusive training dataset. Second, the model is estimated only for a sample of the data and we do not recover cell fixed effects estimates beyond this sample. Third, for the simple DiD approach, we use relative time-to-treatment. This is only available for the matched data, not beyond. For the purpose of accuracy comparison, we therefore construct a simple deforestation model on the training dataset (i.e., it has no information on PA cells after treatment). Since this restricts the available time period included for PA cells, we do not use cell fixed effects, but apply the Mundlak device and include averages of the time varying variables as additional covariates. As a proof of concept, Table ?? shows that we are able to reproduce the average treatment effect of the econometric FE DiD model with this approach. However, it should be noted that the Mundlak device model captures less variation in the outcome variables as indicated by lower R^2 values.

On the training dataset, we then estimate the following model:

$$y_{itc} = \gamma_{tc} + \beta_1 T_i + \beta_2 R_{itc} + \beta_3 \bar{R}_{itc} + \epsilon_{itc}$$

where y_{itc} is the deforestation outcome for grid cell i in year t located in country c from the observations included in the training split in the matched data. γ_{tc} are year-by-country fixed effects. T_i indicates membership of the treatment group, i.e. cells that eventually become protected. R_{itc} is time-varying potential agricultural revenue and $\bar{R}_{itc}$ its mean over the covered time period. ϵ_{itc} is the error term. We run a DiD comparison model including cell fixed effects for comparison. Table L1 shows all four model in comparison, our treatment effect model based on the original sample for the econometric FE DiD approach and the Mundlak device comparison, as well as the simple deforestation model on the training data.

We use this model for prediction on the test data split over the full time period (1986-2020), identically for treated and untreated cells. $\bar{R}_{itc}$ is therefore created over the full time period. This deliberately differs from the training estimation that contains no post-treatment information.

On the dataset used for estimation (matched data, train split), average predicted deforestation aligns well with observed deforestation on average. Higher average deforestation levels in the matched test split and in the full data over the longer time period result in underpredicted average deforestation based on this econometric Mundlak model (see Tables L2 and L3).

Both models, the DL model and the econometric DiD Mundlak model estimated on the training data, are compared on the identical held-out test set of 462,216 cell-years using the same metric definitions. Because the DiD Mundlak model is linear and not bounded at zero deforestation values, for this comparison a cell is classified as positive when its predicted value exceeds zero, and negative predictions are

Table L1: Comparison econometric FE DiD models versus Mundlak device

	DiD FE (train)	DiD Mundlak (train)	DiD FE (main)	DiD Mundlak (main)
revenue	0.0024** (0.0009)	0.0025** (0.0009)	0.0009* (0.0004)	0.0009* (0.0005)
PA		-0.0008** (0.0002)	0.0988 (1259.8942)	-0.0004* (0.0002)
mean(revenue)		0.0062*** (0.0013)		0.0066*** (0.0013)
after			0.0009*** (0.0003)	0.0011** (0.0004)
after:PA			-0.0013*** (0.0003)	-0.0013*** (0.0003)
mean(after)				-0.0011 (0.0026)
Year $\times$ Country FE	Yes	Yes	Yes	Yes
Gridcell FE	Yes	No	Yes	No
Mundlak controls	No	Yes	No	Yes
Num. obs.	15903724	15903724	10035861	10035861
FE: year-country	159	159	151	151
FE: gridcell	1203322		519151	
R ² (full model)	0.1579	0.0114	0.1399	0.0110
R ² (proj model)	0.0002	0.0070	0.0005	0.0086
Adj. R ² (full model)	0.0890	0.0113	0.0929	0.0110
Adj. R ² (proj model)	0.0002	0.0070	0.0005	0.0086

Note: Standard errors in parentheses. *** indicates significance at the 1% confidence level.

Table L2: Prediction performance of the Mundlak model by data set and split

Data	Split	N	RMSE	MAE	R ²	Mean obs. / pred. deforestation
Matched data	Train	15,903,724	0.0159	0.0029	0.0114	0.0014 / 0.0014
	Test	1,966,706	0.0220	0.0040	0.0132	0.0026 / 0.0018
Full data	Train	118,667,062	0.0246	0.0054	0.0028	0.0036 / 0.0023
	Test	18,382,743	0.0283	0.0063	0.0009	0.0046 / 0.0024

Note: Mundlak model estimated on untreated observations of the matched training split. Matched data: matched grid cells, ± 10 years around the hypothetical PA year, relative year -1 dropped. Full data: all grid cells of the split (matched or not), all years. RMSE = root mean squared error, MAE = mean absolute error. Deforestation is the annual share of the grid cell deforested.

Table L3: Prediction performance of the Mundlak model by data set, split and group

Data	Split	Group	N	RMSE	MAE	R^2	Mean obs. / pred. deforestation
Matched data	Train	All	15,903,724	0.0159	0.0029	0.0114	0.0014 / 0.0014
		Control	9,472,045	0.0183	0.0036	0.0128	0.0019 / 0.0019
		Treated (pre)	6,431,679	0.0117	0.0018	0.0009	0.0008 / 0.0008
	Test	All	1,966,706	0.0220	0.0040	0.0132	0.0026 / 0.0018
		Control	1,352,249	0.0246	0.0049	0.0123	0.0033 / 0.0021
		Treated (pre)	614,457	0.0146	0.0021	0.0031	0.0011 / 0.0011
Full data	Train	All	118,667,062	0.0246	0.0054	0.0028	0.0036 / 0.0023
		Control	103,135,152	0.0260	0.0059	0.0012	0.0040 / 0.0026
		Treated (pre)	15,531,910	0.0113	0.0020	-0.0096	0.0009 / 0.0006
	Test	All	18,382,743	0.0283	0.0063	0.0009	0.0046 / 0.0024
		Control	16,793,908	0.0295	0.0067	-0.0006	0.0050 / 0.0026
		Treated (pre)	1,588,835	0.0110	0.0020	-0.0069	0.0009 / 0.0009

Note: Mundlak model estimated on untreated observations of the matched training split. Matched data: matched grid cells, ± 10 years around the hypothetical PA year, relative year -1 dropped. Full data: all grid cells of the split (matched or not), all years. All: untreated observations only (control cells and treated cells before PA establishment). RMSE = root mean squared error, MAE = mean absolute error.

set to zero before the regression metrics are computed. Table L4 reports the results. The deep learning model performs better on every metric except recall, where the econometric model’s marginal advantage follows from classifying nearly every cell as positive.

Table L4: Performance of the deep learning and econometric models on the same held-out test set.

Model	Confusion matrix (%)				Classification				
	TN	FP	FN	TP	Precision	Recall	F1	Accuracy	ROC AUC
Deep learning	34.74	9.73	2.42	53.10	0.845	0.956	0.897	0.878	0.963
Econometric	5.59	38.89	2.30	53.23	0.578	0.959	0.721	0.588	0.723

Model	Regression, 1 km				Regression, 10 km				Aggregate	
	RMSE	MAE	R^2	Corr	RMSE	MAE	R^2	Corr	Predicted (km ²)	Share
Deep learning	4.74%	1.48%	0.18	0.44	0.27%	0.06%	0.55	0.74	6,505	89.6%
Econometric	5.36%	1.54%	-0.05	0.18	0.39%	0.09%	0.05	0.57	1,320	18.2%

Note: Both models are evaluated on the same 462,216 cell-years, aggregated to 75,643 blocks at 10 km. Cells are classified as positive at a probability of 0.5 for the deep learning model and at a predicted value above zero for the econometric model. RMSE and MAE are percentages of cell area and so are not comparable across resolutions.

Appendix M Autoregressive bias correction of counterfactual predictions

When generating counterfactual deforestation trajectories, the DL model is applied in two inference modes. In the *non-autoregressive* (non-AR) mode, the model receives observed historical data as input for every prediction year. In the *autoregressive* (AR) mode, each year’s predicted deforestation updates the available forest cover and the deforestation value for that year, which are used as inputs for the subsequent year, producing a fully self-contained counterfactual trajectory. AR mode is necessary for counterfactual estimation because post-treatment observations do not exist. But the AR mode introduces compounding error where small systematic biases in individual predictions accumulate across the rollout horizon, causing the trajectory to drift from true values over time.

We correct this with a single zero-preserving procedure applied independently to each inference mode. A random forest (RF) regressor first removes conditional bias, and a per-relative-year multiplicative factor then restores aggregate calibration. The RF is fit only on cells with a non-zero raw prediction,

and separate correctors are estimated for the AR and non-AR predictions.

The calibration set combines pseudo-PA (PPA) training-split pixels across the full temporal range ($t \in [-10, +10]$ relative to treatment) with pre-treatment cells from actual PA pixels ($t < 0$ relative to designation). PPA pixels are unprotected locations matched to actual PA pixels by propensity score; PA pre-treatment cells carry no treatment effect, and including them anchors the corrected counterfactual to the PA pre-implementation trend. Post-treatment PA cells ($t \geq 0$), whose counterfactual is being estimated, never enter the fit, which preserves causal identification. The PPA train/test partition follows the same grid-level 100 km split used for the DL model, so no cell appears in both the fit and the holdout.

For each pixel-year the target is the bias $b_{i,t} = \hat{y}_{i,t} - y_{i,t}$, where $\hat{y}_{i,t}$ is the DL model prediction and $y_{i,t}$ the observed deforestation. The RF learns this bias from four features: the model’s own prediction $\hat{y}_{i,t}$, which captures magnitude-dependent bias; the year relative to protection, which marks how far into the autoregressive rollout a prediction falls and therefore captures compounding error; the calendar year, which captures secular trends in prediction quality; and the country code (Bolivia=0, Brazil=1, Colombia=2, Ecuador=3, Peru=4). The de-biased value subtracts the predicted bias, is clipped at zero, and leaves predicted zeros unchanged,

$$\tilde{y}_{i,t} = \max(0, \hat{y}_{i,t} - \hat{b}_{i,t}) \mathbb{I}[\hat{y}_{i,t} > 0].$$

Because the zero-mass penalty L_{ZM} already trains the model to output exact zeros where no deforestation occurs, preserving these zeros keeps the correction consistent with that design.

The additive step alone slightly underpredicts aggregate deforestation, because it cannot recover loss in cells the model missed. We therefore apply a per-relative-year recalibration that scales the non-zero corrected mass without altering zero cells. For each relative year r , a factor

$$s_r = \text{clip}\left(\frac{\sum_i y_{i,r}}{\sum_i \tilde{y}_{i,r}}, 0, 10\right)$$

is estimated on the fit set (set to 1 where the denominator is zero, and bounded above to prevent blow-up where a relative-year base sum is near zero), giving the final corrected value

$$\hat{y}_{i,t}^{\text{corrected}} = s_r \tilde{y}_{i,t} \mathbb{I}[\hat{y}_{i,t} > 0].$$

The two hard constraints therefore hold by construction: no negative predictions, and predicted zeros remain exactly zero. The correction operates entirely in real (descaled) units. RF hyperparameters were 300 trees, maximum depth 15, and minimum leaf size of 50 observations, with a fixed random seed for reproducibility.

On held-out PPA test grids, the correction restores aggregate calibration: the corrected sum ratio is 1.03 in AR mode and 1.04 in non-AR mode, compared with a raw AR sum ratio of about 3.3 (the uncorrected autoregressive trajectory over-predicts the basin total roughly threefold), while non-AR R^2 is preserved. We note that a sum ratio near one is the target only on control (PPA) cells; on PA cells the corrected counterfactual is expected to exceed the observed total, since that excess is precisely the avoided-deforestation signal rather than a calibration error. The corrected PA predictions are then used in the downstream difference-in-differences analysis.

Appendix N DINOv3 Embeddings Ablation Analysis

We evaluate the contribution of satellite image embeddings derived from the DINOv3 foundation model using a controlled ablation experiment. The goal of this analysis is to isolate the effect of incorporating image-based spatial context on model training dynamics and predictive performance, relative to an otherwise identical architecture that relies exclusively on historical deforestation time series.

Specifically, we compare two model variants. The baseline model uses only the temporal deforestation features as input to both the binary classification head and the regression head. The augmented model adds a parallel input branch that ingests precomputed DINOv3 satellite embeddings, which are fused with the temporal features at the intermediate representation level. All remaining components of the architecture, including network depth, hidden dimensions, loss functions, optimization settings, training schedule, and data splits, are held constant between the two models. This design ensures that any observed differences can be attributed to the inclusion of the image embeddings rather than to changes in model capacity or training procedure.

Figure N1 compares the training and validation learning curves for both model variants. For the binary classification task, which predicts the presence or absence of deforestation, the model augmented

with DINOv3 embeddings converges more rapidly and achieves consistently lower binary cross-entropy loss on both the training and validation sets. A similar pattern is observed for the regression task, where the hybrid loss decreases faster and stabilizes at a markedly lower value when embeddings are included. In both cases, the gap between training and validation curves remains stable, suggesting that the performance gains are not driven by overfitting but by improved feature representation.

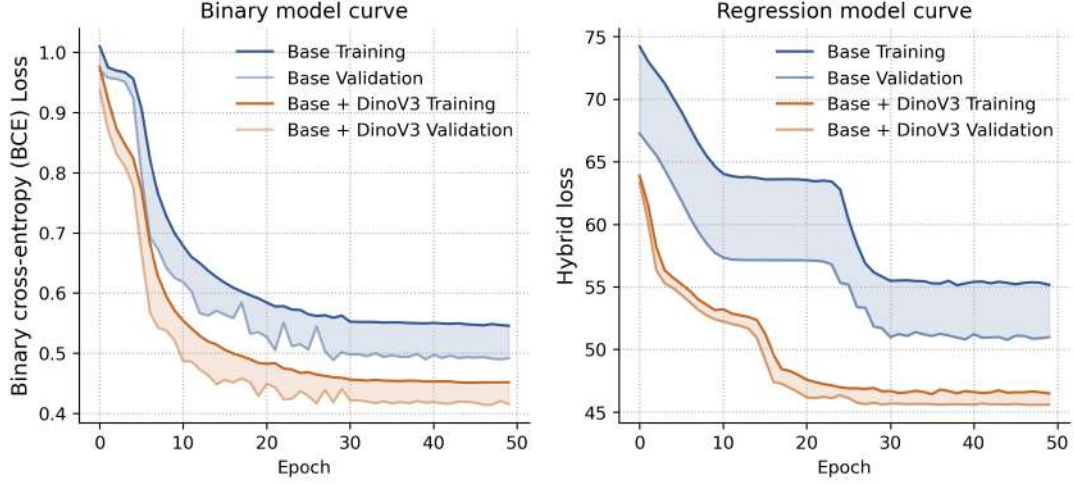


Figure N1: Training and validation loss curves for the baseline model and the baseline model augmented with DINOv3 satellite embeddings, shown separately for the binary classification and regression components.

Beyond training dynamics, we assess both models on a held-out test set to quantify differences in predictive accuracy. Figure N2 summarizes standard evaluation metrics for the regression task, including RMSE, MAE, median absolute error, R^2 , and Pearson correlation. Across all reported metrics, the model augmented with DINOv3 embeddings exhibits improved performance, with lower error magnitudes and higher goodness-of-fit statistics. These results indicate that the image embeddings provide complementary information to the temporal deforestation signals, improving generalization to unseen regions and time periods.

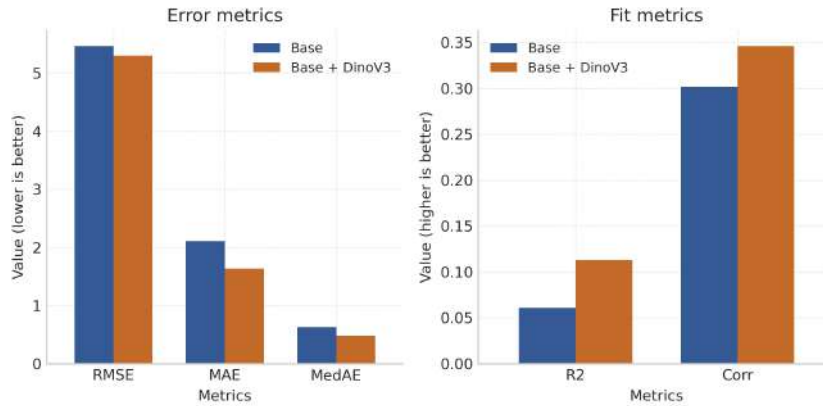


Figure N2: Comparison of test-set performance metrics for the baseline model and the baseline model augmented with DINOv3 satellite embeddings.

Although the absolute gains in evaluation metrics appear limited in magnitude, the consistent reduction in validation loss across training epochs, observed for both the binary and regression objectives that optimize different loss functions, suggests that the DINOv3 embeddings add stable and generalizable signal rather than introducing marginal or noise-driven improvements.

Overall, this ablation analysis demonstrates that incorporating foundation-model-based satellite embeddings yields consistent gains during training and evaluation, supporting their inclusion in the full modeling framework presented in the main text.

Appendix O Spatial heterogeneity in predicted counterfactual deforestation

To assess whether the deep learning model captures the spatial distribution of forest clearing consistently over time, we generate basin-wide predictions for four non-overlapping five-year periods: 1996–2000, 2001–2005, 2006–2010 and 2011–2015. Five-year periods keep the number of eligible grid cells large: a single run spanning the full two decades would have to exclude every grid cell protected at any point across that span, leaving a small and unrepresentative remainder.

Within each period, we exclude grid cells protected at any time up to one year after the period ends, and show them in gray. The one-year margin guards against anticipation effects ahead of designation. Every grid cell shown in color was therefore unprotected throughout the period it appears in. Fewer grid cells remain in each successive period because protected areas continued to be designated over these two decades, so the totals are not directly comparable across periods.

Figure O1 shows observed and predicted forest clearing for each period, summed at 10 km resolution on a colour scale truncated at 20 km² per block and shared across all panels. Predictions are non-autoregressive and bias-corrected. Table O1 reports the corresponding totals.

Table O1: Observed and predicted forest clearing by five-year period and grid cell group, summed over the grid cells shown in Figure O1. Grid cells already protected at the start of each period are excluded. Matched pseudo-protected areas are the unprotected grid cells paired to protected areas in the matching procedure; the remaining grid cells are those outside the matched sample.

Period	Group	Grid cells	Observed (km ²)	Predicted (km ²)	Difference (km ²)	Ratio
1996–2000	Protected after period	1,017,628	3,218	3,193	–25	0.99
	Matched pseudo-PA	881,472	7,251	6,914	–337	0.95
	Outside matched sample	3,706,312	111,790	91,110	–20,680	0.82
	All	5,605,412	122,260	101,217	–21,043	0.83
2001–2005	Protected after period	438,193	1,423	1,898	475	1.33
	Matched pseudo-PA	881,472	11,874	11,930	56	1.00
	Outside matched sample	3,706,312	114,327	103,200	–11,127	0.90
	All	5,025,977	127,625	117,028	–10,597	0.92
2006–2010	Protected after period	176,156	551	674	123	1.22
	Matched pseudo-PA	881,472	10,575	10,623	48	1.00
	Outside matched sample	3,706,312	65,920	68,326	2,406	1.04
	All	4,763,940	77,046	79,623	2,577	1.03
2011–2015	Protected after period	75,452	299	290	–9	0.97
	Matched pseudo-PA	881,472	9,462	9,444	–18	1.00
	Outside matched sample	3,706,312	49,161	49,044	–117	1.00
	All	4,663,236	58,923	58,778	–145	1.00
2016–2020	Protected after period	11,014	4	11	7	3.08
	Matched pseudo-PA	881,472	16,465	15,587	–878	0.95
	Outside matched sample	3,706,312	60,740	61,531	791	1.01
	All	4,598,798	77,208	77,129	–79	1.00

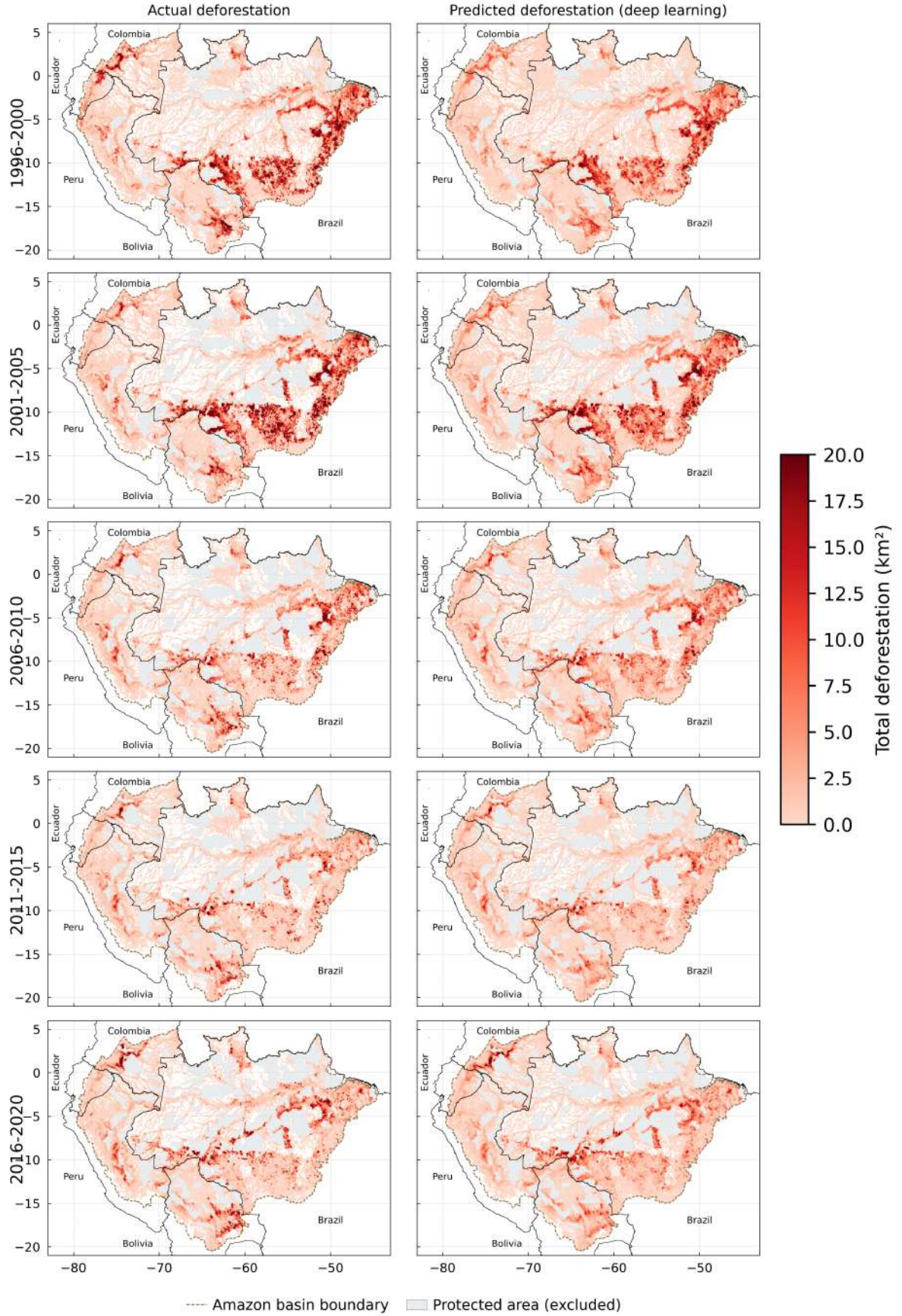


Figure O1: Observed (left) and predicted (right) forest clearing across the Amazon basin for four non-overlapping five-year periods. Grid cells excluded from a period are shown in gray.

Appendix P Error analysis

Appendix Q Econometric estimations

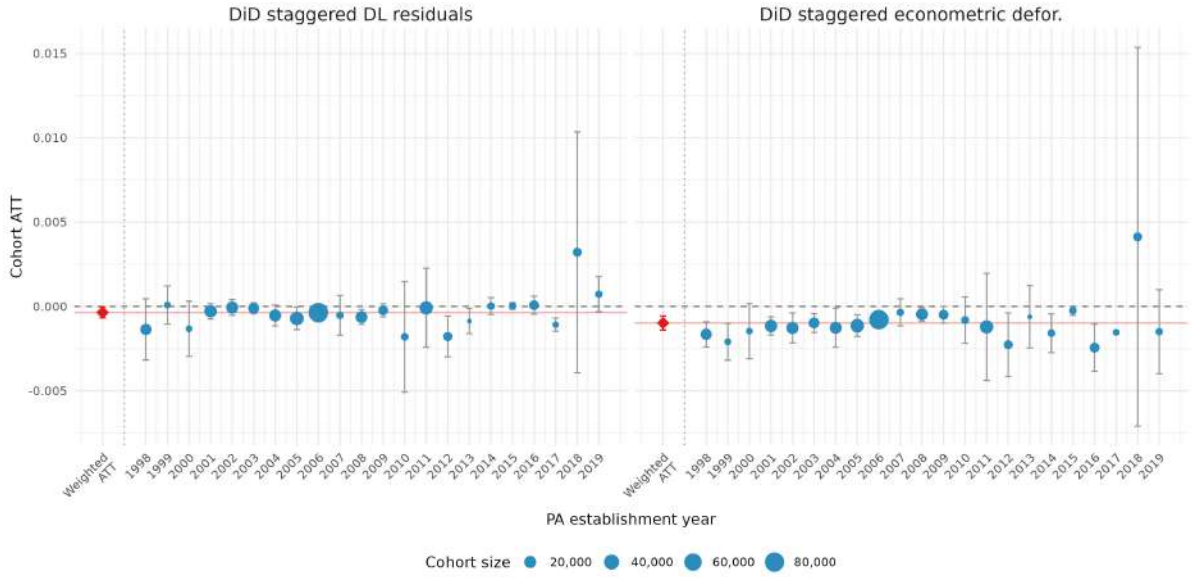


Figure Q1: Cohort-specific estimates of staggered DiD models based on DL residuals (left) and deformation in the pure econometric estimation (right)

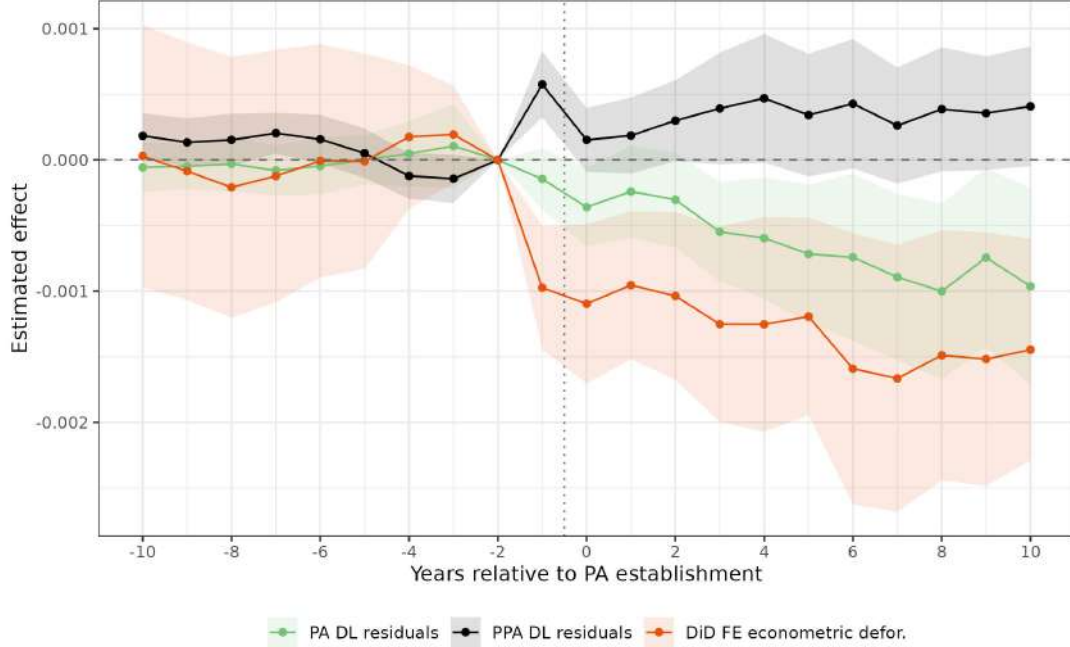


Figure Q2: Event-study plots for non-staggered models, complementary to Figure 3.

Table Q1: Estimation results for DiD models

	PA DL residuals	PPA DL residuals	DiD DL residuals	DiD FE DL residuals	DiD staggered DL residuals	DiD FE econometric defor.	DiD staggered econometric defor.
after implementation	-0.0005*** (0.0002)	0.0003* (0.0002)					
after			0.0002 (0.0001)	0.0003* (0.0002)		0.0009*** (0.0003)	
PA			0.0001** (0.0000)			0.0987 (1226.1000)	
after:PA			-0.0006*** (0.0002)	-0.0005** (0.0002)		-0.0013*** (0.0003)	
cohort after:PA					-0.0004* (0.0002)		-0.0010*** (0.0002)
revenue						0.0009* (0.0004)	0.0024*** (0.0003)
Year × Country FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Grid cell FE	No	No	No	No	No	No	No
Num. obs.	5450820	3718959	9169779	9169779	9169779	10035866	10035866
FE: year-country	124	120	125	125	125	156	156
R ² (full model)	0.0020	0.0002	0.0007	0.0993	0.0993	0.1399	0.1399
R ² (proj model)	0.0002	0.0000	0.0001	0.0001	0.0001	0.0005	0.0005
Adj. R ² (full model)	0.0019	0.0002	0.0007	0.0453	0.0453	0.0929	0.0929
Adj. R ² (proj model)	0.0002	0.0000	0.0001	0.0001	0.0001	0.0005	0.0005
FE: gridcells				519151	519151	519151	519151
Num. obs. (dataset)					9169779		10035866
Num. cohorts					22		22
Num. treated cells					305999		305999

Note: Standard errors in parentheses. *** indicates significance at the 1% confidence level. Defor.=Deforestation, DiD=Difference-in-Differences

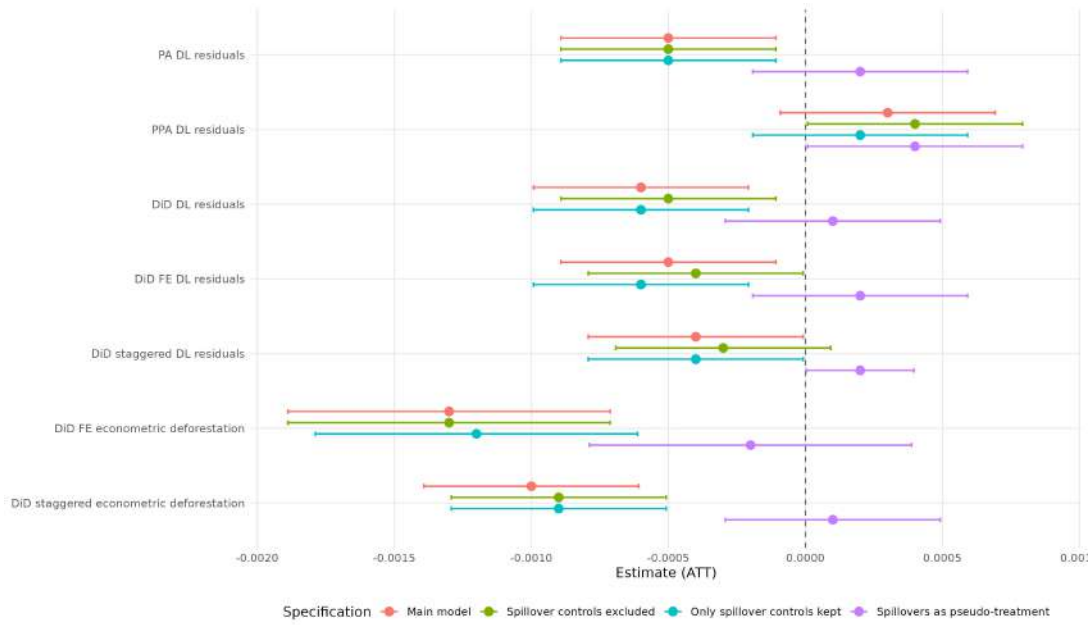


Figure Q3: Robustness tests

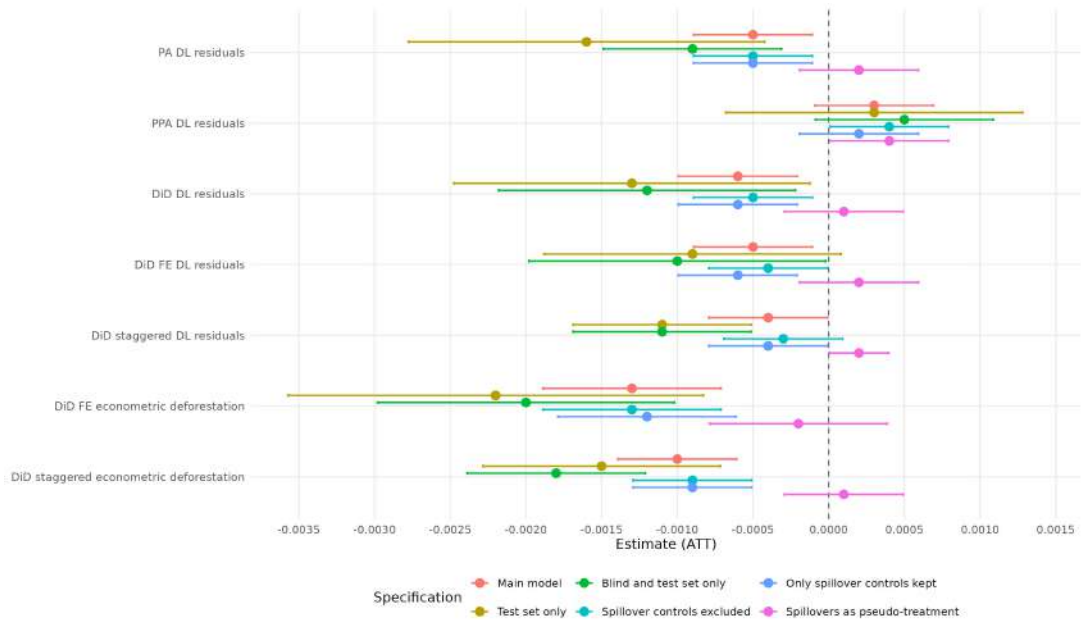


Figure Q4: Robustness tests

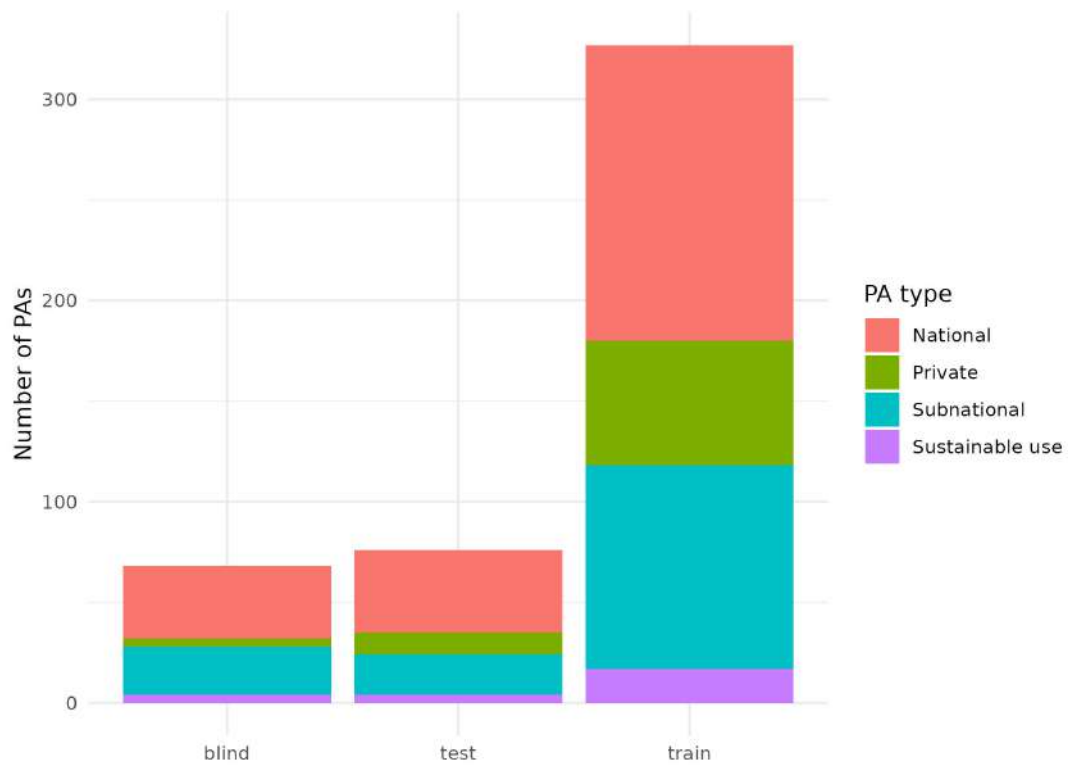


Figure Q5: PA types by sample split in estimation sample

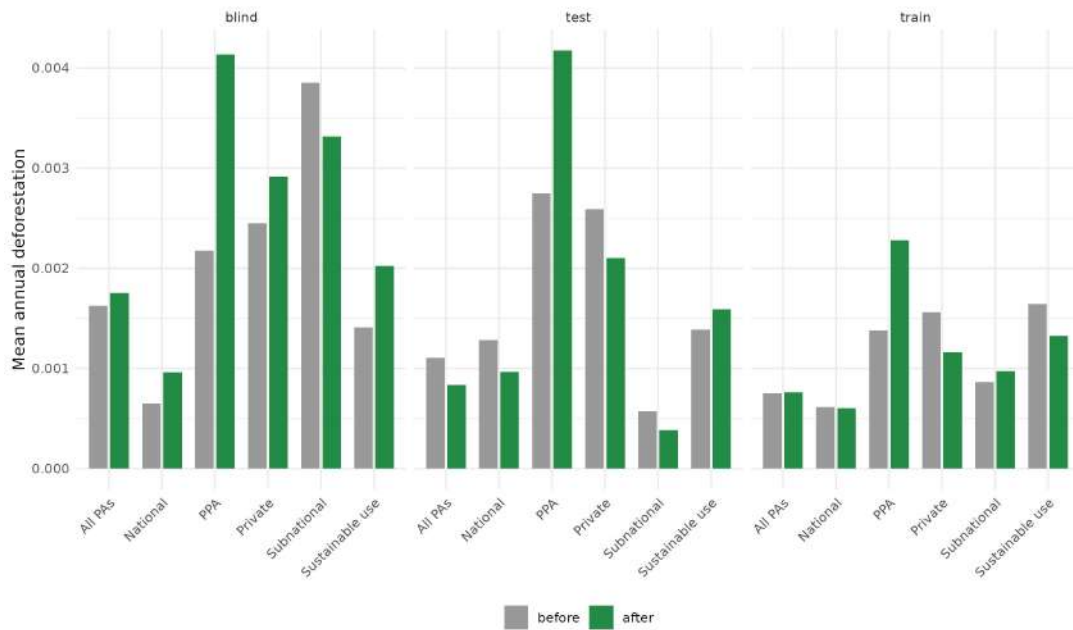


Figure Q6: Deforestation before and after (pseudo-) protection by split in estimation sample and PA type

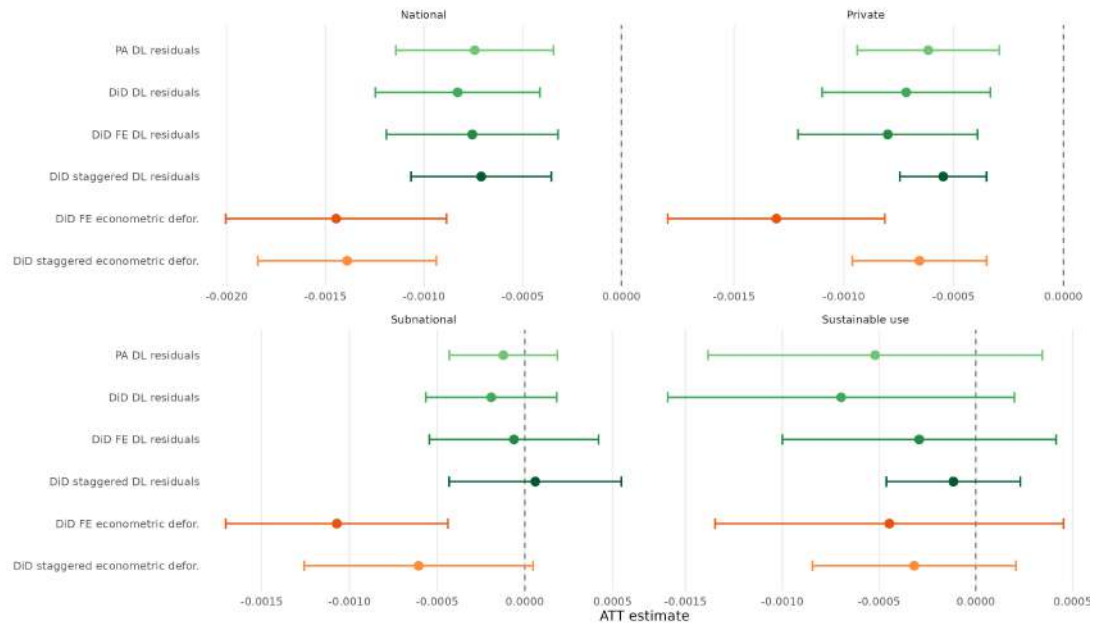


Figure Q7: Estimation results by PA type